

Adaptive Factor Allocation via Machine Learning Regime Detection

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Version : 11 January 2026

Abstract

Equity factor payoffs are strongly time-varying, raising the practical question of whether factor exposure can be adapted across market regimes using a simple, pre-specified, implementable rule rather than discretionary timing. We study adaptive factor allocation via machine-learning-based regime classification in a long-only U.S. equity setting. Using 25 fundamental characteristics, we construct investable signal portfolios and aggregate them into two tradable sleeves: an Offensive basket (growth-oriented signals) and a Defensive basket (quality and conservative signals). Market regimes are defined from an EWMA volatility proxy computed on an investable benchmark and mapped into a binary high/low state. We train multiple supervised classifiers under a common feature set and evaluate them in a walk-forward design with annual refits at June formation dates and a fixed July–June holding horizon. The out-of-sample evaluation spans July 1980–June 2024 and enforces strict point-in-time constraints and explicit timing alignment to preclude look-ahead bias. Performance is reported gross and net of turnover-scaled transaction costs, and risk-adjusted results are assessed under alternative factor models with HAC inference and multiple-testing diagnostics. We find that several regime-switching configurations improve net performance relative to the market benchmark with an economically modest gross-to-net wedge under annual implementation. However, incremental gains relative to transparent buy-and-hold allocations (including a 50/50 Offensive/Defensive benchmark) are typically modest, benchmark-dependent, and sensitive to the risk-adjustment model. Overall, machine learning can add value for regime-based factor allocation in some configurations, but the effect is heterogeneous and implementation frictions are first-order for net performance.

Keywords: machine learning; regime detection; factor investing; asset allocation; volatility regimes; long-only portfolios; transaction costs; out-of-sample evaluation.

JEL subject classification: G11; G12; C45; C53.

Acknowledgments: Maria-Pia Lombardo, Anna Smirnova, Prof. Simon Scheidegger, and Prof. Amit Goyal.

1 Introduction

Equity factor strategies are often presented as long-run sources of return, yet their short- and medium-horizon performance is highly time-varying. Periods of strong factor payoffs can be interrupted by sharp and persistent drawdowns, and the relative attractiveness of “offensive” (growth- and momentum-tilted) versus “defensive” (quality- and conservatism-tilted) exposures appears to shift with market conditions. For long-only investors, this creates a practical allocation problem: static weights across factor sleeves can be easy to implement, but they may leave substantial performance on the table when the market environment changes. The natural question is whether a transparent, investable rule can adapt factor exposure across regimes without relying on ex-post narrative or discretionary timing.

Machine learning offers a tempting answer. In principle, flexible classifiers can combine multiple signals and accommodate non-linearities to detect changes in market state. In practice, however, most “ML timing” claims face two challenges that are especially acute in empirical asset pricing. First, regime detection is vulnerable to subtle lookahead and alignment errors because labels, features, and portfolio formation dates live on different calendars; small timing mistakes can generate large apparent gains. Second, when many model classes, hyperparameters, and feature choices are explored, performance dispersion is inevitable, and ex-post selection can overstate the economic value of the approach. As a result, the central empirical task is not only to build a regime-timing strategy, but to *measure the incremental contribution of machine learning* under a design that is explicit about timing, comparable across models, and transparent about uncertainty.

This paper studies adaptive factor allocation through machine-learning-based regime detection under a deliberately conservative and documented backtesting protocol. We begin with a set of 25 firm-characteristics signals and aggregate them into two investable sleeves: an *Offensive* basket that combines growth-oriented signals and a *Defensive* basket that combines quality and conservative signals (Appendix B reports the full compositions). The allocation problem is cast as a binary regime choice that determines whether the portfolio leans Offensive or Defensive over the next 12 months. Regimes are defined using an EWMA volatility proxy computed from a broad, investable benchmark return series, and models are trained to classify the upcoming high/low volatility state using information available at the forecast date. Portfolio weights are updated annually in June, consistent with common factor-portfolio formation conventions, and each annual decision produces a single out-of-sample holding-period return sequence from July through June.

Our empirical analysis is organized around three questions. First, does ML-based regime switching improve *net* performance relative to transparent buy-and-hold alternatives once conservative transaction costs are applied? Second, are any gains robust to standard risk adjustments and inference procedures (including heteroskedasticity-and-autocorrelation-consistent standard errors), or do they largely reflect systematic exposures that are absorbed under alternative factor models? Third, how concentrated are results across model classes and portfolio constructions—do improvements appear broadly, or only for a subset of specifications—and what does that imply for the practical value of ML in this setting?

To answer these questions, we enforce a strict timing convention throughout the pipeline and evaluate all strategies on an identical out-of-sample period. Signals are formed using point-in-time accounting data and investability filters; betas and features are lagged to ensure that all information sets correspond to what an investor would have observed at the decision date. A long warmup and buffer period is used to initialize rolling estimates and avoid early-sample instability. Table 4 summarizes the complete timing protocol across stages (signal formation, basket construction, feature/label creation, and model training), and shows how warmup windows, lags, and forecast horizons align. The out-of-sample evaluation spans July 1980 through June 2024, with annual refits and a fixed 12-month forecast horizon, so that each model faces the same information set and the same realizations.

We deliberately treat the machine learning component as *an empirical object to be evaluated*, not as a source of guaranteed outperformance. Accordingly, we compare multiple classifier families under a common protocol and report results in a way that emphasizes dispersion rather than a single “best” number. The main analysis considers five basket constructions and five ML model classes (25 strategies in total). Performance is reported both gross and net of transaction costs, and we quantify incremental value relative to transparent benchmarks (including a fixed 50/50 Offensive/Defensive allocation) to isolate the contribution of timing. Risk-adjusted performance is assessed under alternative factor specifications, and the analysis highlights where inference is sensitive to the chosen model. To address data-snooping concerns, we report cross-model distributions and complementary multiple-testing adjustments in the Appendix.

The results point to a qualified conclusion. Risk-adjusted alphas are often positive across specifications, but statistical support is heterogeneous and economically meaningful improvements are concentrated in a subset of strategies. Performance differences are driven jointly by basket construction and model choice: dynamic (time-varying) beta baskets tend to deliver stronger net performance than the fixed economic basket, yet incremental gains relative to transparent buy-and-hold benchmarks are typically modest and sensitive to the risk-adjustment model and the chosen benchmark. Implementability is shaped primarily by turnover and the associated trading frictions, which can erode gross improvements, although under our annual June design the gross-to-net wedge remains economically limited. Taken together, the evidence suggests that machine learning can add incremental value for adaptive factor allocation in some configurations and eras, but that the effect is neither uniform nor automatic.

The remainder of the paper proceeds as follows. Section 2 describes the data sources and the construction of an investable U.S. equity universe. Section 3 details the factor-signal portfolios, basket definitions, volatility-regime labeling, feature construction, and the full training and backtesting protocol. Section 4 reports out-of-sample performance, incremental alpha tests relative to buy-and-hold benchmarks, and robustness across risk models and subperiods. Section 5 discusses interpretation, implementation frictions, and limitations. Section 6 concludes with implications for the practical use of machine learning in regime-based factor allocation.

2 Data

This section details the data sources and the methodology used to construct the investment universe. We focus on U.S. common stocks over the sample period from July 1963 to June 2024, integrating market data with fundamental accounting variables.

To ensure the economic significance and replicability of our factor portfolios, we implement a rigorous data processing pipeline. The remainder of this section is organized as follows: Section 1.1 describes the primary datasets (CRSP and Compustat); Section 1.2 outlines the sample construction, including our strict point-in-time methodology to eliminate look-ahead bias, the stochastic imputation of missing delisting returns, and the liquidity filters applied to define the final investable universe.

2.1 Datasets

Our analysis integrates three primary data sources covering U.S. equity markets. Monthly stock returns are obtained from the Center for Research in Security Prices (CRSP)¹ stock database, annual accounting data from Compustat via Wharton Research Data Services (WRDS)², and the CRSP–Compustat links from the CRSP/Compustat Merged Database (CCM)³. We restrict our main analysis to the period from July 1963 through June 2024.

- **CRSP Monthly Stock File:** Monthly returns, prices, shares outstanding, and exchange codes for all U.S.-listed common stocks (share codes 10 and 11). The dataset spans December 1925 to December 2024, encompassing over 26,000 unique firms across NYSE, AMEX, and NASDAQ exchanges.
- **Compustat Fundamentals Annual:** Annual financial statement data, including balance sheet items, income statement variables, and cash flow metrics. Coverage begins in fiscal year 1960 and extends through fiscal year 2024, with approximately 400 accounting variables per firm-year.
- **CRSP–Compustat Merged (CCM) Link Table:** We merge the CRSP and Compustat datasets using the CRSP–Compustat Merged (CCM) Link Table to match CRSP PERMNOs with Compustat GVKEYs. To ensure a precise one-to-one mapping, we implement a prioritization scheme based on link types.ⁱ
- **Fama–French benchmark factors (FF5/FF6/FF7):** We obtain monthly Fama–French factor returns (Mkt–RF, SMB, HML, RMW, CMA) and the risk-free rate (RF; one-month Treasury bill) from Kenneth French’s data library⁴. We use RF to express all test-asset returns in excess form and to interpret Mkt–RF. We augment FF5 with the momentum factor (UMD) from the same source to form FF6^{FrenchMomentum}, and further add the Quality Minus Junk factor (QMJ) from AQR to form FF7⁵. These factors serve as controls in our asset-pricing tests (Section 4.2).
- **Quality Minus Junk (QMJ):** Monthly QMJ factor returns from AQR’s data library⁵. We use QMJ as an additional “quality” factor to extend the FF6 specification to FF7 in robustness tests (Section 4.2).
- **Q-Factor Model (Q5):** Monthly factor returns (R_MKT, R_ME, R_IA, R_ROE, R_EG) from the augmented q^5 factor model^{6,7}, downloaded from the Global q-Factor Data Library.ⁱⁱ The dataset spans January 1967 through December 2024 (696 months), providing an alternative factor specification based on investment and profitability for our asset-pricing tests (Section 4). The five factors capture market (R_MKT), size (R_ME), investment (R_IA), profitability (R_ROE), and expected growth (R_EG) premia.

2.2 Sample Construction

2.2.1 Point-in-Time Methodology

To avoid look-ahead bias, we implement a strict point-in-time approach closely aligned with Fama and French’s (1993)⁸ and Asness, Moskowitz, and Pedersen’s (2013)⁹. Portfolios are formed annually in June of year t using only data that would have been available to investors at that time. Specifically:

1. **Fiscal Year Window:** For portfolios formed in June of year t , we use the most recent fiscal year-end with a closing

ⁱSpecifically, we prioritise the ‘LC’ (Link Research) link type, which indicates a link that has been manually researched and verified by CRSP. Other link types such as ‘LU’ (Link Unresearched, generated by computer algorithms), ‘LS’ (Link Secondary), or ‘LX’ (Link Exchange) are only considered when a primary verified link is unavailable, or are excluded to prevent matching errors and ensure the primary security is used.

ⁱⁱAvailable at <https://global-q.org/factors.html>

date between January 1 and December 31 of year $t - 1$. This implies that, at portfolio formation, fundamentals are between 6 and 17 months old.

2. **Publication Delay:** We assume that annual accounting data for fiscal year $t - 1$ become publicly available within four months of the fiscal year-end. Since portfolios are formed in June of year t , this implies that all fiscal-year $t - 1$ statements are observable at the formation date, so no forward-looking information is used.
3. **Maximum Staleness:** To prevent excessive data staleness, observations for which the most recent fiscal year-end precedes January 1 of year $t - 1$ (age > 17 months in June of year t) are excluded from signal calculations.

This conservative approach guarantees that all fundamental ratios computed at portfolio formation date reflect only historical information, thereby eliminating forward-looking information leakage.

2.2.2 Delisting Treatment

A well-documented challenge in long-horizon return studies is the treatment of delisted securities.¹⁰ CRSP provides delisting returns (DLRET) for most delistings, but approximately 30% of performance-related delistings (mainly liquidations and non-compliance removals) have missing delisting returns. Shumway (1997)¹⁰ shows that ignoring these missing delisting returns induces a substantial upward bias in average returns, because delistings for poor performance are typically associated with large negative final returns.

Motivated by this evidence, we correct missing performance-related delisting returns using a stochastic imputation scheme designed to mimic the distribution of observed delisting returns:

- **Bankruptcy Delistings** (DLSTCD 400–490):

To account for exchange-specific recovery rates consistent with Shumway (1997), we differentiate by listing venue:

- *NYSE/AMEX*: Impute a delisting return drawn from $\mathcal{N}(-30\%, 15\%)$, reflecting higher average recovery rates.
- *NASDAQ*: Impute a delisting return drawn from $\mathcal{N}(-55\%, 15\%)$, reflecting more severe losses.

All draws are truncated to $[-95\%, -5\%]$ to rule out implausible values.

- **Non-Compliance Delistings** (DLSTCD 500–590):

Impute a delisting return drawn from $\mathcal{N}(-30\%, 10\%)$, truncated to $[-60\%, -5\%]$, regardless of the exchange.

The means and truncation ranges are chosen to produce loss magnitudes consistent with the severe underperformance documented for performance-related delistings¹⁰, while ruling out implausible values. To ensure reproducibility of the stochastic imputation, we employ firm-specific random seeds based on permanent identifiers (PERMNO). The adjusted return for delisting month t is then computed as:

$$r_{i,t}^{\text{adj}} = (1 + r_{i,t}) \times (1 + \text{DLRET}_{i,t}) - 1,$$

where $r_{i,t}$ is the regular monthly return and $\text{DLRET}_{i,t}$ is either the observed or imputed delisting return. Following imputation, we apply a “zombie filter” to remove all post-delisting observations from the panel. This ensures that firms exit the sample permanently upon fatal delisting, consistent with the economic reality of bankruptcy or exchange removal.

2.2.3 Sample Filters

We apply the following sequential filters to construct our final analysis sample:

1. **Common Stock Filter:** Retain only ordinary common shares (CRSP share codes 10 and 11), excluding REITs, ADRs, closed-end funds, and other non-equity securities.
2. **Exchange Filter:** Restrict to NYSE (EXCHCD = 1), AMEX (EXCHCD = 2), and NASDAQ (EXCHCD = 3) (see Table 1).
3. **Price Filter:** Exclude stocks priced below \$5 at portfolio formation to mitigate microstructure noise and bid-ask bounce effects prevalent among penny stocks.
4. **Market Capitalisation Filter:** To mitigate microcap-driven illiquidity and extreme-return effects, we exclude stocks with market equity below the 20th percentile of NYSE firms at the portfolio formation date. This “all-but-microcaps” filter uses the NYSE size breakpoints employed by Fama and French (2008)¹¹ and subsequent anomaly-replication studies.
5. **Data Completeness Filter:** We apply signal-specific data requirements following Fama and French (1993)⁸. For each fundamental signal, we retain only firm-years with non-missing accounting inputs required to construct that signal. As shown in Appendix A.2.1 Table 17, coverage rates range from 61% (working capital accruals) to 95%

(net share issuance) of the eligible universe, with an average of 76% across 25 signals (1,791 firms per June formation) relative to the screened universe (Table 1). This filtering reflects realistic investment constraints under which investors can only trade on observable signals. While data missingness may not be random and could introduce selection effects, this approach mirrors actual trading feasibility.

Table 1: Sample Construction and Filtering Procedure

Period	Years	Raw CRSP	After Exchange	After NYSE size filter	Eligible $\geq \$5$
Early Period	1963–1979	3,045	2,975	1,167	1,152
Middle Period	1980–1999	6,799	6,645	2,435	2,391
Recent Period	2000–2024	6,088	5,960	3,278	3,128
Overall	1963–2024	5,483	5,363	2,427	2,348

Notes: This table reports the average number of firms at each stage of the sample construction process across three distinct periods. *Raw CRSP* includes all securities in the CRSP database. *After Exchange* restricts the sample to NYSE, AMEX, and NASDAQ listings ($EXCHCD \in \{1, 2, 3\}$). After NYSE size filter (20th percentile) applies the Fama-French size screen, requiring market capitalization to exceed the NYSE 20th percentile breakpoint for NYSE listed firms (market equity measured at the end of May and portfolios formed in June). *Eligible $\geq \$5$* further excludes stocks with prices below \$5. Formation dates occur annually in June following Fama and French (1993). The filtering procedure removes micro-cap stocks, penny stocks, and illiquid securities to ensure economically meaningful portfolio analysis. Sample period: July 1963 to June 2024.

Table 1 reports the impact of each filter on sample size across three distinct periods. On average, these filters reduce the raw CRSP universe from 5,483 firms to 2,348 eligible firms known in June of each year (1963–2024). The market capitalisation filter alone removes approximately 55% of firms, ensuring our results are representative of an institutional investment universe rather than theoretical inefficiencies concentrated in illiquid micro-cap stocks.

By imposing these filters, we consciously sacrifice cross-sectional breadth to ensure our results are representative of an institutional investment universe. Our objective is to test anomalies that are potentially tradeable by large asset managers, rather than theoretical inefficiencies concentrated in micro-cap stocks that are costly to trade.

3 Methodology

This section describes the complete empirical design used to evaluate machine-learning-based regime detection for adaptive factor allocation. Our methodology follows a modular pipeline: we first construct investable long-only signal portfolios from firm characteristics, then aggregate these signals into Offensive and Defensive baskets under several basket-construction schemes, and finally apply supervised classifiers to predict the volatility regime that governs the annual allocation between sleeves. Throughout, we impose a strict information-set discipline: decisions implemented at time t use only information available up to $t - 1$, and all formation dates, lags, warmup windows, and out-of-sample boundaries are explicitly documented to preclude look-ahead bias.

The remainder of the section is organized as follows. Section 3.1 details the construction of characteristic-sorted signal portfolios (June formation, July–June holding period) and the long-only implementation. Section 3.2 defines the set of 25 fundamental signals and their grouping. Section 3.3 introduces five basket-construction strategies—one fixed economic mapping and four dynamic beta-based mappings—and describes how basket returns are formed. Section 3.4 defines the economic and classification metrics used to evaluate the resulting strategies, while Section 3.5 presents the econometric framework for risk-adjusted performance (alphas) and inference. Section 3.6 specifies the machine learning setup (labels, features, and model classes). Section 3.7 consolidates the timing convention and walk-forward backtesting protocol, including the warmup/buffer structure and the common out-of-sample window shared across models. Finally, Section 3.8 describes the transaction cost model used to translate gross performance into net, implementable returns.

3.1 Portfolio Construction

3.1.1 Signal Definition and Sorting

Following Fama and French’s (1993)⁸, we employ NYSE breakpoints to define decile portfolios, thereby avoiding the well-known small-stock bias that arises when using full-sample breakpoints. Specifically, in June of each year t , using signal values based on fiscal year $t - 1$ fundamentals and market equity measured at the end of May (i.e., $ME_{i,t} = P_{i,\text{May}(t)} \times \text{SHROUT}_{i,\text{May}(t)}$), we:

1. Rank all NYSE stocks by signal s and compute the 10th and 90th percentiles;
2. Apply these NYSE-based breakpoints to the full universe (NYSE + AMEX + NASDAQ) to assign stocks to deciles;
3. Form portfolios P1 (bottom 10%, low signal) through P10 (top 10%, high signal);
4. Construct long-only portfolios by taking the relevant extreme decile (P10 or P1) according to the signal’s theoretical

direction.

3.1.2 Weighting Scheme and Rebalancing

Portfolios are formed annually in June of year t and held through June of year $t + 1$. Consistent with our objective to capture the pure factor signal independent of size biases, we focus exclusively on equal-weighted (EW) returns. At the formation date, the weight of each stock i in portfolio P is defined as:

$$w_{i,t} = \frac{1}{N_{P,t}}$$

where $N_{P,t}$ is the number of firms in the portfolio at formation.

Since equal-weighting requires rebalancing contrarian to price movements, we rigorously model the associated trading frictions and turnover in Section 3.8 (Transaction Cost Modeling).

3.1.3 Investment Universe Filters

To ensure sufficient ex-ante diversification, we require a minimum of 10 firms per decile portfolio at the formation date (June). If a specific signal fails to meet this breadth requirement in a given year, the entire year is excluded for that signal. Post-formation attrition is handled via our delisting return methodology, accepting that portfolio breadth may naturally decrease during the holding period.

Portfolio backtests span 732 monthly observations (July 1963–June 2024), with portfolios formed annually in June using accounting data from fiscal year-ends at least six months prior. The final complete portfolio was formed in June 2023, with returns observed through June 2024.

The full implementation calendar and information-set alignment are summarized in 3.7.1.

3.1.4 Historical Data Requirements – Eligibility Criteria

Following Fama and French (1992, 1993)¹², we impose a minimum historical return requirement for portfolio inclusion. Specifically, firms must have at least 24 months of non-missing returns in the period preceding portfolio formation.

To accommodate this requirement while allowing for sporadic missing data, we employ a 30-month lookback window. For portfolios formed in June of year t , we collect monthly returns from December of year $t-3$ through June of year t (30 months total). Firms with fewer than 24 valid observations within this window are excluded from portfolio formation.

This approach provides a 6-month buffer (20% tolerance) to account for temporary trading suspensions, data gaps, or other sporadic missing observations, while maintaining the strict minimum of 24 months of data.

3.2 Fundamental Signals

We construct 25 fundamental signals from annual Compustat data, grouped into five primary economic dimensions consistent with our summary statistics (Table 17): Profitability, Investment, Accruals & Quality, Liquidity & Solvency, and Operating Efficiency. Our signal set follows standard constructions commonly used in the empirical factor literature and aligns with the characteristic definitions catalogued in the factor zoo (Feng, Giglio, and Xiu, 2020)¹³; Table 15 reports the precise formulas, Compustat mnemonics, and Zoo IDs. Each signal is computed using only historical data available at the portfolio formation date (June of year t), strictly adhering to the point-in-time constraints described in Section 2.2.1.

This comprehensive selection is guided by two strategic objectives tailored to our long-only investment framework:

1. **Economic Heterogeneity:** By spanning the entire financial statement spectrum, we minimise multicollinearity in the feature set. This ensures that the machine learning algorithm receives complementary sources of risk and alpha, rather than redundant information.
2. **Dynamic Rotation Potential:** Our investment strategy is strictly long-only and relies on rotating capital across a broad set of 25 characteristic-sorted signal portfolios. This breadth provides sufficient cross-sectional variation for the model to identify which signal characteristics deliver superior risk-adjusted performance across market regimes.

To mitigate the influence of extreme outliers, all signals are winsorised cross-sectionally at the 1% and 99% levels each year. Detailed definitions for each signal are provided in Appendix A.1.

Finally, we assess the independence of our information set. As reported in Appendix A.2.2 (Figures 8 and 9), the average pairwise *rank* correlation across signals is low (0.091), suggesting limited redundancy in the cross-section. For diagnostic purposes, we also compute correlations between the monthly return spreads of signal-implied long–short portfolios (P10–P1), whose average correlation is 0.110.

Analysing these long–short spread correlations is informative because differencing out common return components provides a clean diagnostic of overlap across characteristic premiums.ⁱⁱⁱ For a long-only investor, this helps identify signals with low overlap and motivates diversified allocations across the corresponding long legs (e.g., P10 portfolios) without requiring short positions. The wide dispersion of spread correlations (down to -0.830) indicates substantial heterogeneity in the dynamics of the signal-implied premiums, which is valuable for the learning algorithm when reallocating risk across regimes.

3.3 Portfolio Construction Strategies

To test the robustness of our machine learning framework across different portfolio construction methodologies, we implement five distinct basket classification strategies. Each strategy employs the same 25 fundamental signals but differs in how factors are allocated between Offensive (high-risk, high-return) and Defensive (low-risk, stable) portfolios. This multi-strategy approach allows us to disentangle the predictive power of machine learning from the specific portfolio construction method.

3.3.1 Basket construction and implementation

3.3.1.1 Economic Classification (Basket 1)

The first strategy employs a **fixed economic classification** based on the underlying fundamentals of each signal. We manually assign each of the 25 signals to either the Offensive or Defensive basket based on theoretical priors about their risk-return characteristics;

Signal Assignment. Table 19 reports the full mapping. The **Offensive basket** (9 signals) collects growth- and expansion-related measures (e.g., sales growth, capital expenditure, and turnover improvements) that are typically associated with stronger exposure to the business cycle. The **Defensive basket** (16 signals) groups profitability, quality, liquidity, and balance-sheet strength measures (e.g., gross profitability, cash holdings, leverage, and accruals) that are commonly viewed as indicators of financial resilience. This classification is fixed ex ante and held constant throughout the sample.

3.3.1.2 Dynamic Beta Classification (Baskets 2–5)

The remaining four strategies employ dynamic allocation based on time-varying systematic risk, measured by the market beta $\beta_{s,t}$. Unlike the fixed economic classification, these strategies reassign factors *monthly* between High-Beta (offensive) and Low-Beta (defensive) baskets based on CAPM slopes estimated from regressions of factor excess returns on the excess return of our investable-market benchmark (equal-weighted over the screened liquid universe). Each month, we rank all 25 factors by $\beta_{s,t}$ and allocate the top 50% (12–13 factors) to the High-Beta basket and the bottom 50% to the Low-Beta basket.

Throughout, we estimate betas using monthly excess returns: $r_{s,t}^e = r_{s,t} - r_{f,t}$ and $r_{mkt,t}^e = r_{mkt,t} - r_{f,t}$, where $r_{f,t}$ is the Fama–French risk-free rate (RF) and $r_{mkt,t}$ is the return on our investable-market benchmark (equal-weighted NYSE20 universe with a \$5 minimum price filter).

Beta Estimation Methodologies. We estimate time-varying factor betas using four distinct econometric approaches:

1. Basket 2 – Expanding Window Beta (July 1968–June 2024):

$$\beta_{s,t} = \frac{\text{Cov}(r_s^e, r_{mkt}^e)_{[\text{Jul } 1963: t-1]}}{\text{Var}(r_{mkt}^e)_{[\text{Jul } 1963: t-1]}}$$

Uses all available history from inception to month t . Requires minimum 60 months (5 years) for initial estimation. First valid portfolio: **July 30, 1968**.

2. Basket 3 – Rolling Window Beta, 60-Month (July 1968–June 2024):

$$\beta_{s,t} = \frac{\text{Cov}(r_s^e, r_{mkt}^e)_{[t-59: t-1]}}{\text{Var}(r_{mkt}^e)_{[t-59: t-1]}}$$

ⁱⁱⁱThese long–short spreads are used for diagnostics only and are not intended as implementable strategies in our setting.

Uses a fixed 5-year trailing window. More reactive to regime changes but exhibits “cliff effects” when old observations drop out. Requires 60 months. First valid portfolio: **July 30, 1968**.

3. **Basket 4 – EWMA Beta, $\lambda = 0.94$ (July 1968–June 2024):**

$$\beta_{s,t} = \frac{\sum_{\tau=1}^{t-1} \lambda^{(t-1)-\tau} r_{s,\tau}^e r_{\text{mkt},\tau}^e}{\sum_{\tau=1}^{t-1} \lambda^{(t-1)-\tau} \left(r_{\text{mkt},\tau}^e\right)^2}, \quad \lambda = 0.94$$

Applies exponential decay with $\lambda = 0.94$, yielding half-life of 11.2 months. Recent observations receive higher weight, providing smooth adaptation without cliff effects. Requires 60 months initialization. First valid portfolio: July 30, 1968.

4. **Basket 5 – Kalman Filter Beta (August 1963–June 2024):**

$$\begin{aligned} \text{State Equation: } \beta_{s,t} &= \beta_{s,t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, Q) \\ \text{Observation Equation: } r_{s,t}^e &= \beta_{s,t} r_{\text{mkt},t}^e + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, R) \end{aligned}$$

Employs a Kalman filter to estimate time-varying betas under a random-walk state equation¹⁴. We set $Q = 0.0001$ and estimate R from OLS residuals. The filter can be initialized after the first return observation; we nevertheless apply the one-month information lag described in Section 3.7.2. First valid portfolio: **August 30, 1963**.

Monthly basket formation, the one-month lag, and warmup/buffer periods are detailed in 3.7.2.

3.3.2 Basket Construction and Portfolio Mechanics

3.3.2.1 Portfolio Formation and Timing

We form 25 long-only signal portfolios from annual June sorts and hold their stock-level constituents from July year y through June year $y+1$, yielding a monthly return series with no intra-year rebalancing at the holdings level (July 1963–June 2024). Basket 1 applies a fixed economic mapping of these signal portfolios into Offensive and Defensive groups. For Baskets 2–5, basket composition is updated monthly by reclassifying signals between High Beta and Low Beta sleeves based on lagged market-beta estimates (formed using information up to $t-1$ and implemented in month t). Regime-based allocation between the Offensive and Defensive sleeves is applied at the annual decision date and held over the subsequent July–June period (see Timing Convention in Section 3.7).

3.3.2.2 Weighting and Benchmark

Within each basket, we employ equal-weighting across constituent signals to avoid concentration risk. For each signal, we construct equal-weighted decile portfolios using NYSE breakpoints (P10 for “high is good” signals, P1 for “low is good” signals), then aggregate basket returns as the equal-weighted average of their respective signal portfolios.

We define the *NYSE Benchmark* as an equal-weighted portfolio of all stocks above the NYSE 20th-percentile market-cap breakpoint (ex-microcap; “NYSE20”), with a \$5 minimum price screen. The NYSE Benchmark is reconstituted annually (using market equity measured at the end of May) and held from July of year t through June of year $t+1$. This benchmark excludes microcap stocks while maintaining broad market exposure over the full 1963–2024 sample period.

3.3.3 Summary of Basket Strategies

Table 2 summarizes the key characteristics of all five strategies. Basket 1 provides a theory-driven baseline with low turnover (annual rebalancing only). Baskets 2–5 offer increasing degrees of adaptability: Expanding Beta is the most stable, Rolling Beta the most reactive, EWMA balances smoothness and responsiveness, and the Kalman filter provides a model-based state-space estimate of time-varying betas. By comparing machine learning performance across these heterogeneous construction methods, we reduce the likelihood that results are driven by a particular beta-estimation choice or portfolio construction convention.

Table 2: Summary of Basket Construction Strategies

Basket	Method	Rebalancing	Beta Window	Key Feature
1	Economic Classification	Annual (June)	N/A	Fixed theory-based allocation
2	Expanding Beta	Monthly	Inception– t	Stable, slow adaptation
3	Rolling Beta (60m)	Monthly	$[t - 59 : t]$	Reactive, cliff effects
4	EWMA Beta ($\lambda = 0.94$)	Monthly	Exponential decay	Smooth, half-life 11m
5	Kalman Filter Beta	Monthly	State-space model	Model-based state-space estimate

Notes: This table summarizes the five portfolio construction strategies used to evaluate machine learning regime-switching performance. *Basket 1* employs a fixed classification based on economic intuition (Table 19). *Baskets 2–5* dynamically allocate factors monthly based on time-varying market beta estimates, differing in estimation methodology. All strategies use equal-weighting within baskets, NYSE breakpoints for decile sorts, and a one-month lag to prevent look-ahead bias. The market benchmark is the equal-weighted portfolio of all eligible common stocks in our screened universe (stocks above the NYSE 20th-percentile size breakpoint and with price $P_t \geq \$5$).

3.4 Performance Metrics

3.4.1 Overview

We evaluate model performance along two complementary dimensions: (i) *economic performance* of the resulting long-only strategy (returns, risk, and risk-adjusted performance), and (ii) *classification performance* of the regime forecasts that determine the annual sleeve allocation. Unless stated otherwise, all economic metrics are computed from monthly out-of-sample returns, and all classification metrics are computed exclusively on out-of-sample regime predictions over the common evaluation window defined in Section 3.7.4.

3.4.2 Economic performance (portfolio-level)

Let $r_{p,t}$ denote the monthly gross return of strategy/portfolio p in month t and let $r_{f,t}$ denote the Fama–French risk-free rate (RF). We work with excess returns $r_{p,t}^e = r_{p,t} - r_{f,t}$ when reporting Sharpe ratios and risk-adjusted performance.

Core performance and risk. For each model and basket, we report annualised mean return, annualised volatility, and the Sharpe ratio computed from monthly data:

$$\mu_{\text{ann}} = 12 \cdot \bar{r}, \quad \sigma_{\text{ann}} = \sqrt{12} \cdot \sigma(r), \quad \text{SR}_{\text{ann}} = \sqrt{12} \cdot \frac{\bar{r}^e}{\sigma(r^e)},$$

where $\bar{r}$ denotes the sample mean of monthly returns and $\sigma(\cdot)$ the sample standard deviation. We also report the annualised Sortino ratio,

$$\text{SoR}_{\text{ann}} = \sqrt{12} \cdot \frac{\bar{r}^e}{\sigma^-(r^e)},$$

where $\sigma^-(r^e)$ denotes the standard deviation of negative excess returns. We additionally report maximum drawdown, skewness, and kurtosis as complementary downside and tail-risk diagnostics.

Gross vs. net performance. We report both gross and net returns. Net returns deduct the transaction cost model described in Section 3.8, applied consistently with the realised turnover implied by the underlying rebalancing rules (signal-level and basket-level). When netting returns, we recompute all economic performance statistics (e.g., Sharpe, drawdown) using the resulting net return series.

Risk-adjusted performance (alphas). We evaluate abnormal performance via time-series regressions of portfolio excess returns on benchmark factor models:

$$r_{p,t}^e = \alpha_p + \beta_p^\top \mathbf{f}_t + \varepsilon_{p,t},$$

where $\mathbf{f}_t$ is the factor vector (FF5/FF6/FF7, M4, or Q5 depending on the specification in the Results section). Statistical significance of α_p is assessed using HAC standard errors (Newey–West) computed on monthly data.

Multiple testing. Because we report alphas across multiple baskets, models, and factor specifications, we also report multiple-testing adjustments using a false discovery rate (FDR) procedure (Benjamini–Hochberg), alongside raw p -values.

3.4.3 Classification performance (regime forecasts)

Let $y_t \in \{0, 1\}$ denote the realised regime at decision time t (e.g., LOW vs. HIGH volatility), and let $\hat{y}_t$ denote the out-of-sample predicted regime produced by a model. When models output probabilities $\hat{p}_t = \mathbb{P}(\hat{y}_t = 1)$, we evaluate both threshold-based classification performance and probability-based discrimination.

Threshold-based metrics. Based on the confusion matrix counts (TP, FP, TN, FN) for the HIGH regime, we report:

- **Accuracy:** $\frac{1}{N} \sum_{t=1}^N \mathbb{I}\{\hat{y}_t = y_t\}$
- **Precision (HIGH):** $\frac{TP}{TP+FP}$
- **Recall (HIGH):** $\frac{TP}{TP+FN}$
- **F1-score (HIGH):** $2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

Probability-based metrics. We additionally report:

- **ROC–AUC:** Area under the receiver operating characteristic curve (threshold-free discrimination).
- **Brier score:** $\frac{1}{N} \sum_{t=1}^N (\hat{p}_t - y_t)^2$ (probability calibration).

All classification metrics are computed exclusively on out-of-sample predictions over the common evaluation period in Section 3.7.4.

3.4.4 Return Measurement and Test Statistics

For each fundamental signal s , the portfolio construction in Section 3.1 produces monthly returns for decile portfolios $P1_{s,t}, \dots, P10_{s,t}$. Let $P_{s,t}^L$ and $P_{s,t}^H$ denote, respectively, the “low” and “high” sides of the signal. For signals where higher values are theoretically associated with higher expected returns (e.g., value or profitability), $P_{s,t}^H$ is the top decile (P10) and $P_{s,t}^L$ is the bottom decile (P1). For signals where lower values are theoretically attractive (e.g., some risk or distress measures), the assignment is reversed.

3.4.5 Long-only and long–short return series

Long-only implementation. Our investable strategies are strictly long-only. Accordingly, we retain a single implementable leg per signal,

$$P_{s,t}^{LO} \in \{P_{s,t}^H, P_{s,t}^L\},$$

where $P_{s,t}^{LO} = P_{s,t}^H$ if the high side is theoretically attractive and $P_{s,t}^{LO} = P_{s,t}^L$ otherwise. Unless stated otherwise, performance statistics and portfolio aggregation in subsequent sections are computed on these long-only signal portfolio returns.

Long–short spreads (diagnostics). For benchmarking and diagnostic purposes, we also consider the corresponding decile spread

$$R_{s,t}^{LS} = P_{s,t}^H - P_{s,t}^L,$$

so that a positive $R_{s,t}^{LS}$ always corresponds to the theoretically attractive side of the signal outperforming the unattractive side. These long–short spreads are used to assess signal strength and overlap across characteristics, but they are not treated as implementable strategies in our long-only setting.

3.4.6 Summary statistics and annualisation

We summarise raw performance using the time-series mean return $\bar{R}$, its annualised value $12 \times \bar{R}$, and the monthly standard deviation. Sharpe ratios are computed as the mean monthly excess return divided by its monthly standard deviation and annualised by multiplying by $\sqrt{12}$. Where statistical tests on mean returns (or alphas) are reported, we compute t -statistics using HAC (Newey–West) standard errors on monthly data.

3.5 Econometric Approach

To assess risk-adjusted performance, we estimate time-series factor regressions for each *implementable* long-only portfolio return series. Let $r_{p,t}^e = r_{p,t} - r_{f,t}$ denote the monthly excess return of portfolio p (signal-portfolio, basket, or strategy return), where $r_{f,t}$ is the Fama–French risk-free rate (RF). Our baseline specification employs the Fama–French six-factor model (FF6):

$$r_{p,t}^e = \alpha_p + \beta_p^\top \mathbf{f}_t^{FF6} + \varepsilon_{p,t},$$

where $\mathbf{f}_t^{FF6}$ includes the market excess return (Mkt–RF), size (SMB), value (HML), profitability (RMW), investment (CMA), and momentum (UMD). The intercept α_p (annualised as $12 \times \alpha_p$) measures abnormal performance unexplained by these standard risk premia.

To test whether estimated alphas are robust to alternative factor structures, we additionally report results under the following benchmark models:

1. **Fama–French 6 + Quality (FF7):** We augment FF6 with the *Quality Minus Junk* (QMJ) factor¹⁵. This specification tests whether returns to accounting-based signals are subsumed by a broad quality premium capturing profitability, safety, growth, and payout characteristics.
2. **Hou–Xue–Zhang q^5 model (Q5):** We employ the q^5 model¹⁶, which augments the Hou–Xue–Zhang (2015) q -factor model with an expected growth factor (REG). Grounded in investment-based asset pricing theory, this model provides a stringent benchmark for signals related to investment and fundamental improvement.

For all specifications, statistical inference is based on Newey–West (1987)¹⁶ heteroskedasticity- and autocorrelation-consistent (HAC) standard errors with 12 monthly lags to account for serial correlation induced by annual portfolio formation.

3.6 Machine Learning Framework

3.6.1 Volatility Target and Regime Definition

3.6.1.1 Target Label Construction

To define the ground truth for our supervised learning models, we construct a volatility target based on a robust 5-year rolling window.

EWMA Volatility Target: For each formation date t (June 30), we compute the realized volatility state using the exponentially weighted moving average (EWMA) of the NYSE Benchmark returns. The computation strictly adheres to a 60-month (5-year) backward-looking window, ensuring the target captures the structural volatility environment leading up to formation:

$$\sigma_{\text{EWMA},t}^2 = \lambda \cdot \sigma_{\text{EWMA},t-1}^2 + (1 - \lambda) \cdot r_{m,t}^2, \quad \lambda = 0.94$$

Construction Protocol:

- **Window Size:** Rolling 60 months (5 years) ending in May of year t .
- **Data Usage:** Returns from $t - 60$ months to $t - 1$ month.
- **Lookahead Prevention:** The return of June year t is explicitly excluded from the target calculation at formation.
- **Warm-up Period:** July 1963 – June 1968 (60 months) initialize the EWMA. First target label (Y_t): June 30, 1969 (using returns through May 31, 1969 — 71 months available).

Binary Regime Classification: The continuous EWMA volatility is transformed into a binary regime indicator (Y_t) to serve as the classification label. To prevent lookahead bias, we employ an expanding median threshold computed exclusively on past observations. A regime is classified as "High Volatility" (1) if the current EWMA exceeds the expanding median of all prior EWMA values:

$$Y_{\text{BINARY},t} = \begin{cases} 1 & \text{(HIGH VOLATILITY) if } \sigma_{\text{EWMA},t} > \text{Median} \left(\{ \sigma_{\text{EWMA},\tau} \}_{\tau=1969}^{t-1} \right) \\ 0 & \text{(LOW VOLATILITY) otherwise} \end{cases}$$

Supervised Learning Framework: While Y_t is mechanically computable at formation (using May t returns), the machine learning models serve a distinct purpose: *learning to recognize regime patterns from fundamental and technical features* (X_t). The models (Logistic, Random Forest, XGBoost, LSTM) identify which combinations of momentum, beta dispersion, and cross-sectional characterize High vs. Low volatility regimes, enabling dynamic portfolio rebalancing.

Lookahead control: All predictors and the regime label use information available up to $t - 1$ (see Section 3.7.3, Table 4).

This setup frames the problem as regime recognition: the algorithms learn to detect volatility states from observable market characteristics, rather than mechanically computing EWMA. The value lies in discovering which feature combinations provide the strongest regime signals.

3.6.2 Feature Engineering

We construct 35 features from two sources: (1) individual portfolio returns (50 portfolios: 25 factors $\times$ {P10, P1}), and (2) expanding-window market betas (25 factors). Features span five categories: momentum, dispersion/volatility, beta dynamics, cross-sectional characteristics, and interaction terms.

Table 3: Summary of Model Feature Blocks

Feature block	#	Frequency	Max window	Availability / lag
Momentum	6	Monthly	12 months	Uses returns up to $t-1$
Dispersion & Volatility	8	Monthly	12 months	Uses returns/vol up to $t-1$
Beta Dynamics	7	Monthly	12 months	Betas estimated through $t-1$
Cross-sectional Dist.	8	Monthly	3 months	Computed from cross-section at $t-1$
Interactions	6	Monthly	12 months	Deterministic combinations

Note: All inputs are constructed using information strictly available at time $t-1$. Detailed definitions are reported in Appendix C.1.

3.6.2.1 Feature Taxonomy and Sources

Table 20 (in Appendix) presents the complete feature set with mathematical definitions. All features are cross-sectional aggregates (mean, median, or standard deviation) computed over the 50 characteristic portfolios or 25 beta estimates.

3.6.2.2 Lookahead Prevention (Lag Structure)

Critical Adjustment: All 35 features undergo a +1 month shift to eliminate lookahead bias. This ensures features X_t used for formation at date t rely strictly on the information set $\mathcal{F}_{t-1}$:

$$X_t^{\text{shifted}} = X_{t-1}^{\text{raw}}$$

For example, for a portfolio formation on June 30, 1970, features are computed using data ending in May 1970. The return of June 1970 is explicitly excluded from the input set. For more details, see Section 3.7.3.

3.6.2.3 Multicollinearity and Feature Retention

Because our feature set includes engineered interaction terms and closely related aggregates, some degree of collinearity is expected. Since our objective is out-of-sample prediction rather than coefficient interpretation, we retain the full set of 35 features and rely on model-specific regularisation and validation to mitigate overfitting. In particular, tree-based models are generally robust to correlated predictors, logistic regression is L2-regularised, and the LSTM uses dropout. Model selection and all reported performance are based strictly on out-of-sample evaluation.

3.6.3 Model Specifications

We benchmark four classification algorithms representing distinct machine learning paradigms. All training, validation, and portfolio implementation strictly follow the one-month information lag described in Section 3.7.4. Detailed hyperparameter grids and optimization protocols are provided in Table 21 (Appendix C.2).

3.6.3.1 L2-Regularized Logistic Regression (Baseline)

The probability of a high-volatility regime is modeled as:

$$P(Y_t = 1 \mid X_t) = \frac{1}{1 + \exp\left(-\left(\beta_0 + \sum_{k=1}^{35} \beta_k X_{k,t}\right)\right)}$$

We optimize the coefficients by minimizing the negative log-likelihood with an L2 penalty (Ridge):

$$\mathcal{L}(\beta) = - \sum_{i=1}^N [y_i \log p_i + (1 - y_i) \log(1 - p_i)] + \frac{1}{2C} \|\beta\|_2^2$$

Features are standardized (Z-score) using statistics computed strictly on the training set.

3.6.3.2 Random Forest Classifier

We employ an ensemble of $B = 200$ decision trees with bootstrap aggregation. The prediction is the majority vote of the ensemble:

$$\hat{P}(Y_t = 1 \mid X_t) = \frac{1}{B} \sum_{b=1}^B \mathbb{1}\{\text{Tree}_b(X_t) = 1\}$$

To prevent overfitting given the sample size, we use shallow trees (maximum depth = 3).

3.6.3.3 XGBoost Classifier

We utilize gradient boosting with regularized additive tree learners. The prediction at step m is given by:

$$\hat{y}_t^{(m)} = \hat{y}_t^{(m-1)} + \eta \cdot f_m(X_t)$$

where the learner f_m is chosen to minimize the regularized objective:

$$\mathcal{L}^{(m)} = \sum_{i=1}^N \ell(y_i, \hat{y}_i^{(m)}) + \gamma T + \frac{1}{2} \lambda \|\omega\|^2$$

To address class imbalance (63% LOW / 37% HIGH), the `scale_pos_weight` parameter is dynamically recomputed at each annual formation based on the training set distribution.

3.6.4 LSTM Neural Network

The sequential dependency is modeled via a two-layer LSTM architecture with dropout regularization:

$$\begin{aligned} h_t^{(1)} &= \text{LSTM}_1(X_t, h_{t-1}^{(1)}) \quad (32 \text{ units}) \\ h_t^{(1)} &\leftarrow \text{Dropout}(h_t^{(1)}, p = 0.2) \\ h_t^{(2)} &= \text{LSTM}_2(h_t^{(1)}, h_{t-1}^{(2)}) \quad (32 \text{ units}) \\ \hat{y}_t &= \sigma(W h_t^{(2)} + b) \end{aligned}$$

Input features are reshaped into 3D tensors ($N, 1, 35$). Feature standardization is applied based on training data statistics, which proved critical for convergence in our ablation studies.

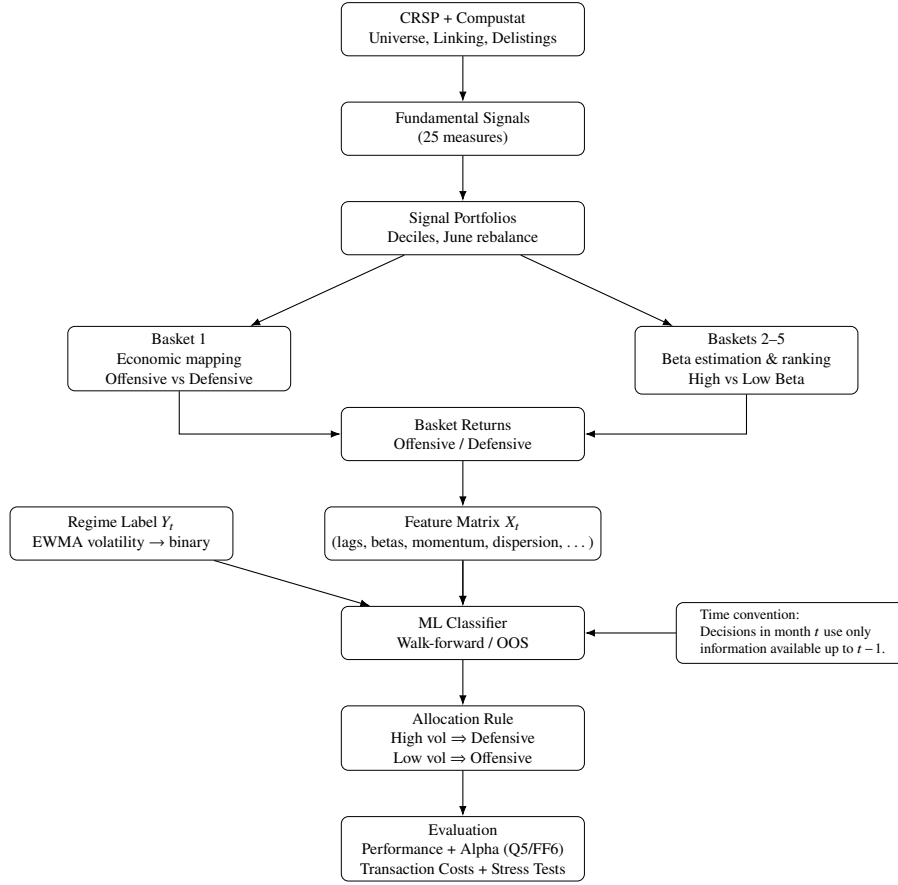
3.7 Timing Convention

Our empirical design relies on a multi-frequency structure combining **annual signal formation (June rebalancing)** with **monthly portfolio returns** and an **annual walk-forward regime forecast** applied over the subsequent July–June holding period. To preclude look-ahead bias, all decisions implemented at time t are functions only of the information set $\mathcal{F}_{t-1}$ available strictly prior to t . This subsection summarises the temporal structure across four stages: (1) signal portfolio formation, (2) basket construction, (3) feature engineering and regime labelling, and (4) machine learning training and backtesting. Figure 1 provides an overview of the dual-frequency structure, while Tables 4 and 5 report the detailed timing protocol and model alignment.

3.7.1 Signal Portfolios: Annual Rebalancing

We form 25 fundamental signal portfolios annually in June. Formation in June year y uses fiscal year $y - 1$ accounting data (see Section 2.2.1 for point-in-time constraints) and market equity measured at the end of May year y . Portfolios are equally-weighted within deciles, ranked by NYSE breakpoints. Each of the 25 signals yields two extreme decile portfolios (P10 and P1), resulting in 50 decile-portfolio return series held from July y through June $y + 1$. Monthly returns are

Figure 1: Macro architecture of the pipeline



Data construction, signal portfolios, basket formation, regime labelling and feature engineering, machine learning classification, allocation, and evaluation.

sourced from CRSP over this 12-month holding period.

For the long-only implementation, we retain one leg per signal (either P10 or P1 depending on the signal's expected direction), yielding 25 long-only signal portfolios used in the subsequent basket construction and beta-based classification.

Timing structure.

- **Formation date:** June 30, year y
- **Information set:** Fundamentals $FY(y-1)$, market equity end-May(y)
- **Holding period:** July(y) – June($y+1$) (12 months, no intra-year rebalancing)
- **Rebalancing frequency:** Annual (June only)

3.7.2 Basket Formation: Monthly Dynamic Beta Classification

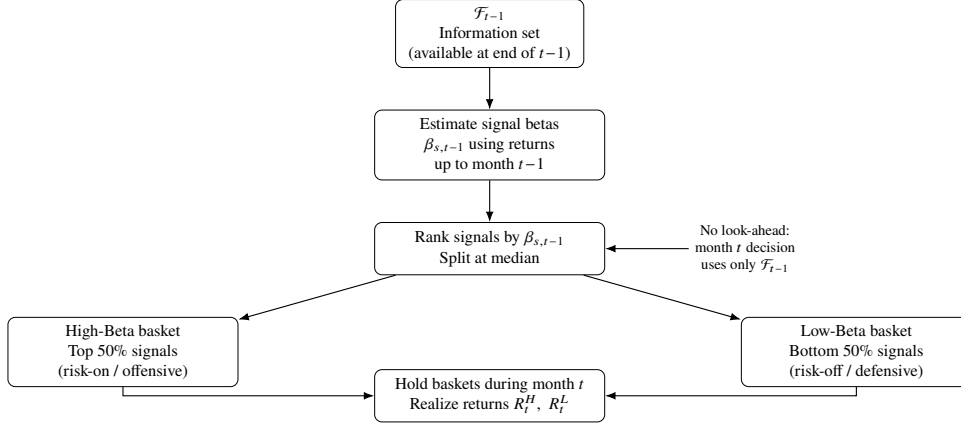
We construct five basket strategies combining the 25 long-only signal portfolios. Basket 1 employs a fixed allocation based on economic intuition (9 Offensive signals, 16 Defensive signals). Baskets 2–5 implement dynamic beta-based classification, allocating signals to High Beta or Low Beta groups monthly.

For dynamic baskets, we estimate monthly market betas $\beta_{s,t}$ for each of the 25 signals using four methods: (i) Expanding window, (ii) Rolling 60-month window, (iii) EWMA with $\lambda = 0.94$ [17], and (iv) Kalman Filter with time-varying states.

Anti-lookahead protocol. At the end of month $t-1$, we rank signals by $\beta_{s,t-1}$ and allocate the top 50% to High Beta, bottom 50% to Low Beta. This allocation is held during month t . Critically, beta $\beta_{s,t}$ (computed using returns through end-of-month t) is *never* used to form month t positions—we implement a systematic 1-month lag via `.shift(1)` operations in code, ensuring formation at t uses $\beta_{s,t-1}$.

Warmup and buffer periods. All beta estimation methods require a 60-month warmup period (July 1963 – June 1968) for stable initialization, yielding first valid estimates in July 1968. An additional buffer period (July 1968 – June 1980) accommodates the ML training warmup (discussed below), ensuring all pipeline components are aligned when out-of-sample backtesting begins in July 1980.

Figure 2: Monthly High/Low Beta Basket Reclassification (No-Lookahead Timing)



Dynamic beta-based basket formation. At the end of month $t-1$, signal betas $\beta_{s,t-1}$ are estimated and used to rank signals. High- and Low-Beta baskets are formed (top/bottom 50%) and held over month t .

3.7.3 Feature Engineering and Regime Labeling: Monthly Alignment

We engineer 35 predictive features organized into five categories: momentum (6 features), cross-sectional dispersion (8 features), beta characteristics (8 features), cross-sectional spreads (6 features), and interaction terms (7 features). Features are computed from the 50 portfolio returns and 25 beta estimates using rolling windows spanning up to 12 months.

Global lag structure. All 35 features undergo a global 1-month forward shift to prevent lookahead. Formally, the feature matrix used at formation t is:

$$X_t^{\text{shifted}} = X_{t-1}^{\text{raw}}$$

ensuring formation in June year y uses features computed through May year y , excluding June returns.

Regime label construction. Our binary classification target derives from EWMA volatility of the market portfolio (NYSE Benchmark), computed with decay parameter $\lambda = 0.94$:

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_t^2$$

The regime label Y_t equals 1 (HIGH volatility) if σ_t^{EWMA} exceeds the expanding median $\text{Median}\{\sigma_\tau\}_{\tau=1969}^{t-1}$, and 0 (LOW volatility) otherwise. The expanding median computed at t uses data strictly before t , combined with the 1-month lag on volatility itself.

Warmup period. EWMA volatility and feature construction require a **60-month warmup** (July 1963 – June 1968), with the first complete feature set and regime label available in June 1969.

3.7.4 ML Training and Out-of-Sample Backtesting

We implement walk-forward validation with annual model retraining. Each June year y , we refit machine learning models on the expanding training set:

$$\text{TrainSet}_y = \{(X_t, Y_t) : t \in [\text{June } 1969, \text{June } (y-1)]\}, \quad X_t \in \mathcal{F}_{\text{May } t}$$

Following annual rebalancing conventions [FamaFrench1993], training sets end in June of year $(y-1)$ for formations in year y , ensuring strict temporal separation between training observations and subsequent holding period returns.

Because X_t is constructed with a one-month lag, the observation indexed at month t uses information available through $t-1$.

The fitted model generates a single out-of-sample prediction for June year y , which determines basket allocation for the subsequent 12-month holding period (July y – June $y + 1$). During this holding period, the model is not retrained; monthly returns are realized without holdings-level rebalancing (annual June reconstitution; July–June holding period), while signal membership may update monthly in Baskets 2–5 via lagged beta reclassification.

Model architectures and training protocols. We evaluate four architectures (Logistic Regression, Random Forest, XGBoost, LSTM) across two training protocols (Expanding window, Rolling 10-year window). The first formation requiring a minimum 10-year training history occurs in June 1980 (training window: June 1969 – June 1979, 121 months). The final formation occurs in June 2023 (training window: June 1969 – June 2022, 637 months for expanding models), generating predictions for July 2023 – June 2024.

Out-of-sample period and buffer. **Out-of-sample backtest:** July 1980 – June 2024 (44 years, 528 months). All five models share this identical evaluation period, ensuring fair performance comparison. The backtest ends in June 2024 because the final annual formation occurred in June 2023; extending beyond June 2024 would require a formation in June 2024, which in turn would demand fiscal year 2023 data plus the requisite publication lag—data not yet available under our point-in-time constraints.

Annual Walk-Forward Expanding-Window Backtesting (June Refit, July–June Forecast Horizon)

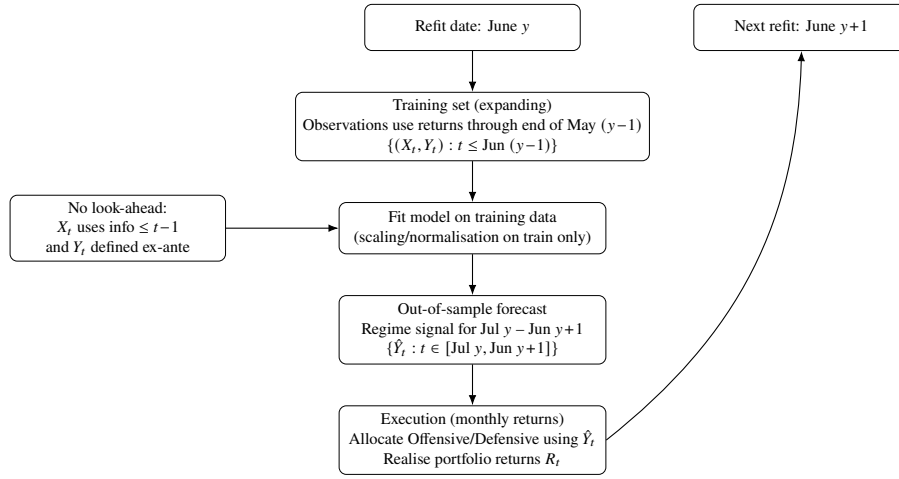


Figure 3: Annual Walk-Forward Expanding-Window Backtesting Protocol. Models are refit annually in June y using training indices up to June $(y - 1)$. Because features incorporate a one-month lag, the last training observation effectively uses returns through May $(y - 1)$. The refitted model then generates an out-of-sample regime forecast implemented over the subsequent July–June holding period (e.g., the June 2010 refit uses information through May 2009 and is evaluated from July 2010 onward).

Warmup: 121 months (June 1969 – June 1979) for minimum 10-year training window. **Buffer:** The period from the first complete feature set (June 1969) to the start of out-of-sample testing (July 1980) serves as a buffer ensuring all pipeline components—signals, betas, features, regime labels—are fully initialized and aligned before performance evaluation begins.

To ensure full transparency and prevent any lookahead bias, Table 4 summarizes the complete timing convention across all stages of the pipeline, including formation dates, information sets, and warmup/buffer windows.

Table 4: Comprehensive Timing Protocol Summary

Component	Frequency	Formation Date	Information Set	Warmup / Buffer / First Valid
Stage 1: Signal Portfolios				
P10/P1 portfolios	Annual	June 30, y	FY($y - 1$), ME(May y)	Jun 1963 (first formation)
Holding period	Monthly	Jul(y) – Jun($y + 1$)	–	12-month rebalancing cycle
Stage 2: Basket Construction				
Basket 1 (Fixed)	Annual	June 30, y	Economic classification	Jun 1963 (no warmup)
Baskets 2–5 (Beta)	Monthly	End of month $t - 1$	$\beta_{s,t-1}$ (lagged)	Warmup: 60m → Jul 1968
Stage 3: Features & Label				
Feature matrix X_t	Monthly	End of month $t - 1$	Returns through $t - 1$	Warmup: 60m → Jun 1969
Regime label Y_t	Monthly	End of month $t - 1$	EWMA vol through $t - 1$	Warmup: 60m → Jun 1969
Stage 4: ML Training & Backtest				
Model refit	Annual	June 30, y	Train: Jun 1969 – Jun($y - 1$)	Warmup: 121m (10Y) → Jun 1980 Buffer: Jun 1969 – Jun 1980
Forecast horizon	12 months	Jul(y) – Jun($y + 1$)	Single prediction per year	Annual forecast
Out-of-sample test	Monthly	Jul 1980 – Jun 2024	528 months, 44 years	OOS period

Note: This table documents timing conventions across all pipeline stages. **Warmup periods** ensure stable estimation: 60 months for beta/EWMA/feature initialization (first valid: Jul 1968 for betas, Jun 1969 for features/label), and 121 months for ML training (first valid: Jun 1980 requiring 10-year history). **Buffer period** (Jun 1969 – Jun 1980) separates warmup completion from out-of-sample testing, ensuring all components are fully aligned. All formation decisions at time t use information sets $\mathcal{F}_{t-1}$, enforced via systematic 1-month lags on betas, features, and regime labels. The out-of-sample period (Jul 1980 – Jun 2024) is identical across all models, ensuring fair comparison.

Building on this timing convention, Table 5 documents the training-window design for each classifier and verifies that all models share an identical out-of-sample evaluation period, ensuring comparability of results.

Table 5: Model Training Protocols and Comparability

Model	Strategy	First Train Start	First Train End	First Size	Last Size	Formations	OOS Period
Logistic Reg.	Expanding	1969-06	1979-06	121 months	637 months	44	Jul 1980 – Jun 2024
Random Forest	Expanding	1969-06	1979-06	121 months	637 months	44	Jul 1980 – Jun 2024
XGBoost	Expanding	1969-06	1979-06	121 months	637 months	44	Jul 1980 – Jun 2024
LSTM-Expanding	Expanding	1969-06	1979-06	121 months	637 months	44	Jul 1980 – Jun 2024
LSTM-Rolling 10Y	Rolling	1970-07	1979-06	120 months	120 months	44	Jul 1980 – Jun 2024

Note: All models share identical out-of-sample testing periods (Jul 1980 – Jun 2024, 528 months, 44 years), ensuring fair performance comparison. Logistic, Random Forest, XGBoost, and LSTM-Expanding employ expanding window protocols with first training spanning Jun 1969 – Jun 1979 (121 months). LSTM-Rolling 10Y uses a fixed 120-month window by design, starting one month later (Jul 1970) to maintain exactly 10 years. This minor 1-month offset (<1% difference in training data) does not affect comparability. All models produce 44 annual formations (1980–2023), with each formation generating predictions for the subsequent 12-month holding period.

3.7.5 Illustrative Example: Formation June 30, 2010

At the June 30, 2010 formation date, signal portfolios are formed using FY2009 accounting data and end-May 2010 market equity, and held from July 2010 to June 2011. Dynamic baskets use lagged beta estimates computed through May 2010 (i.e., $\beta_{s,t-1}$), and the feature vector X_t is constructed using information available through May 2010 due to the global one-month lag. The regime label is computed from EWMA volatility through May 2010 and compared to an expanding median based only on past observations. The model is refit using training data ending strictly before the test date and produces a single out-of-sample prediction for the July 2010–June 2011 holding period.

3.7.6 Verification and Conclusion

We verify temporal alignment via automated implementation checks, ensuring:

- All rolling windows satisfy the required warmup periods (first valid: July 1968 for betas; June 1969 for features/label).
- A global one-month information lag is applied to all features and beta inputs.
- Walk-forward training windows satisfy Train End < Test Date for all 44 formations.
- Fiscal-year accounting data respect point-in-time availability constraints (Section 2.2.1).

Detailed verification protocols are provided in the Online Appendix. This design is intended to preclude look-ahead bias: all decisions at formation t rely exclusively on information sets $\mathcal{F}_{t-1}$ available strictly prior to t .

3.8 Transaction Cost Modeling and Net Returns

While our primary analysis focuses on gross returns to establish the existence and predictability of anomalies, assessing economic significance requires explicit accounting for trading frictions. We compute net returns by deducting estimated

transaction costs from gross portfolio returns, ensuring our findings represent achievable alphas for institutional investors.

3.8.1 Cost Application Framework

Transaction costs are applied following the annual rebalancing convention outlined in Section 3.7. Portfolios are reconstituted once per year in June, when holdings are updated based on the previous fiscal year's financial statements. Our cost framework recognizes that *holdings-level* trading frictions arise at the annual June rebalancing point, as no intra-year rebalancing occurs at the holdings level.

The monthly net return is defined as:

$$r_t^{net} = r_t^{gross} - TC_t^{hold}$$

where the holdings-level transaction cost TC_t^{hold} is non-zero only in June:

$$TC_t^{hold} = \begin{cases} 2 \times \tau_t c_t / 10000 & \text{if } t \text{ is June (rebalancing month)} \\ 0 & \text{otherwise} \end{cases}$$

Here, τ_t denotes portfolio turnover computed on a one-way basis, $\tau_t = \frac{1}{2} \sum |\Delta w|$, and c_t is the per-side (one-way) proportional trading cost rate (in basis points). Net returns apply transaction costs on a round-trip basis (buys and sells), so that the cost deducted in rebalancing months equals $TC_t^{hold} = 2 \tau_t c_t / 10000$.

The holdings-level transaction cost component is applied only at the annual June rebalancing date (and is set to zero in non-rebalancing months). In a sensitivity analysis, we additionally allow for allocation-layer costs TC_t^{alloc} whenever sleeve weights change, which may occur outside June.

In our baseline specification, we model transaction costs only at the holdings level within each long-only signal portfolio. A conservative sensitivity analysis that additionally includes allocation-layer switching costs is reported in Appendix D.2 (Tables 23 and 24).

Transaction costs (holdings-level). Our baseline transaction cost model captures trading in the underlying equities when each long-only signal portfolio is reconstituted at the annual June formation date (June sorts; July–June holding period). Let TC_t^{hold} denote the holdings-level cost, computed as a function of portfolio turnover at the June rebalance and a time-varying per-dollar trading cost c_t (in bps). For monthly accounting, TC_t^{hold} is applied to the June observation and set to zero in non-rebalancing months.

Allocation-layer switching costs (sensitivity). In a sensitivity analysis, we additionally account for allocation-layer trading induced by reallocating capital between the Offensive and Defensive sleeves as signal memberships change over time. Let $w_{s,t}$ denote the weight of signal portfolio s in the sleeve at month t . We define allocation turnover as

$$\tau_t^{alloc} = \frac{1}{2} \sum_{s=1}^{25} |w_{s,t} - w_{s,t-1}|,$$

and set $TC_t^{alloc} = 2 \times c_t \tau_t^{alloc} / 10000$. Net returns are therefore

$$r_t^{net} = r_t^{gross} - TC_t^{hold} \quad (\text{baseline}), \quad r_t^{net} = r_t^{gross} - TC_t^{hold} - TC_t^{alloc} \quad (\text{sensitivity}).$$

3.8.2 Individual Asset Turnover

For each fundamental signal s , we compute a monthly turnover series for its associated long portfolio (P10 or P1, depending on the signal's expected directionality), which is non-zero only at the annual June rebalance under the June formation and July–June holding convention. Turnover for signal s at time t is:

$$\tau_{s,t} = \frac{1}{2} \sum_{i=1}^{N_s} |w_{i,s,t}^{target} - w_{i,s,t}^{drift}|$$

where:

- $w_{i,s,t}^{target} = \frac{1}{N_{s,t}}$ is the target equal weight for stock i in signal s 's portfolio at rebalancing

- $w_{i,s,t}^{\text{drift}}$ is the weight of stock i just before rebalancing, after drifting due to price movements:

$$w_{i,s,t}^{\text{drift}} = \frac{w_{i,s,t-1} \times (1 + R_{i,t-1 \rightarrow t})}{\sum_{j \in P_s} w_{j,s,t-1} \times (1 + R_{j,t-1 \rightarrow t})}$$

- $N_{s,t}$ is the number of stocks in signal s 's portfolio at formation

The factor of $\frac{1}{2}$ in the above equation converts the sum of absolute weight changes into one-way turnover, thereby avoiding double-counting of trades (i.e., buys and sells are recorded once).

3.8.3 Basket-Level Cost Aggregation

Each basket B (e.g., Offensive, Defensive, High Beta, Low Beta) is constructed as an equal-weighted combination of K_B constituent signals. The basket-level turnover is computed as the arithmetic mean of individual signal turnovers:

$$\tau_{B,t} = \frac{1}{K_B} \sum_{s \in B} \tau_{s,t}$$

This aggregation reflects our equal-weight allocation across signals within each basket. For example, the Offensive basket combines 9 growth-oriented signals, while the Defensive basket combines 16 quality and conservative signals (see Appendix B, Table 19 for the full basket compositions). The basket turnover $\tau_{B,t}$ represents the fraction of the basket's portfolio value that must be traded to implement the annual rebalancing.

Transaction costs for basket B at rebalancing time t are then:

$$TC_{B,t} = 2 \times \tau_{B,t} c_t / 10000$$

For the regime-switching strategies presented in Section 3.3.1.2, the final net return aggregates across baskets according to the model's allocation weights.

3.8.4 Time-Varying Cost Schedule

We adopt a conservative one-way (per-side) cost schedule calibrated to institutional trading costs, following Frazzini et al.^{18,19}. We apply round-trip costs at the June rebalancing, using a time-varying cost grid that reflects three distinct market regimes:

Table 6: Per-Side Transaction Cost Schedule and Round-Trip Application

Period	Per-Side Cost c_t (bps)	Market Regime
1980–1999	50	Higher-friction trading environment
2000–2009	30	Post-decimalization transition
2010–2024	20	Modern low-friction era

Notes: This table reports the assumed *per-side* proportional transaction cost rate c_t (in basis points). Turnover is measured on a one-way basis, $\tau_t = \frac{1}{2} \sum |\Delta w|$. Net performance applies *round-trip* costs (buys and sells), so that transaction costs in rebalancing months equal $TC_t = 2 \tau_t c_t / 10000$.

These estimates are *conservative* for our sample for three reasons:

1. **Liquidity screen:** Our NYSE 20th percentile market-capitalization filter excludes the smallest and least liquid stocks, concentrating the sample in large- and mid-cap segments where effective spreads are typically lower than our assumed costs. Lesmond, Ogden, and Trzcinka (1999)²⁰.
2. **Universe composition:** Frazzini, Israel, and Moskowitz (2012)¹⁸ calibrate costs on the full CRSP universe, including micro-caps with bid-ask spreads exceeding 200 bps. Our NYSE-filtered sample systematically excludes these high-friction names. (see Table 19 in Appendix B for full basket compositions).
3. **Cost calibration:** Our assumed costs are intentionally conservative relative to estimates reported for large and liquid U.S. equities. Frazzini, Israel, and Moskowitz (2018)¹⁹.

Therefore, our reported net alphas represent a **lower bound** on achievable performance for institutional investors operating at scale.

3.8.5 Break-Even Transaction Cost Analysis

To assess the economic robustness of our machine learning strategies beyond baseline cost estimates, we compute break-even transaction cost levels—the maximum sustainable trading friction at which excess returns are fully eroded.

For each strategy, we define the headroom as the additional transaction cost capacity (in basis points per year) that would eliminate alpha relative to a benchmark:

$$\text{Headroom} = (r^{\text{net}} - r^{\text{bench}}) \times 10,000$$

where r^{net} is the strategy's annual net return and r^{bench} is the benchmark return.

The **total transaction cost at break-even** represents the maximum sustainable cost level:

$$\text{TC}_{\text{BE}} = \max(0, \text{TC}_{\text{actual}} + \text{Headroom})$$

where $\text{TC}_{\text{actual}}$ denotes realized transaction costs under our baseline model. The max operator ensures economic interpretability: negative headroom (underperformance) implies zero sustainable additional costs.

The transaction cost multiplier quantifies robustness as a ratio:

$$\text{TC Multiplier} = \frac{\text{TC}_{\text{BE}}}{\text{TC}_{\text{actual}}}$$

A multiplier exceeding 1.0 indicates the strategy retains positive alpha even if transaction costs rise above current levels. For instance, a multiplier of 3.0 implies costs can triple before alpha vanishes.

We compute break-even metrics against two benchmarks:

1. **Market benchmark:** The NYSE Benchmark (our primary passive alternative).
2. **50/50 benchmark:** A static 50% Offensive / 50% Defensive portfolio, rebalanced annually in June with transaction costs computed using cost schedule (see Table 6).

This dual-benchmark framework isolates (i) alpha versus passive market exposure and (ii) alpha attributable to dynamic timing relative to a diversified long-only alternative. Full break-even results are reported in the Appendix.

4 Results

This section reports the out-of-sample empirical evidence on the economic value of machine-learning-based regime switching for adaptive factor allocation. We first document implementable performance relative to transparent benchmarks, distinguishing *signal power* (gross returns) from *realizability* (net returns after transaction costs). We then evaluate whether the observed outperformance reflects compensation for systematic risk exposures or residual abnormal performance using standard factor-model regressions. We also assess stability across time and provide predictive diagnostics to interpret the quality of the underlying volatility regime signal that drives the allocation rule.

The remainder of the section is organized as follows. Section 4.1 reports out-of-sample performance results, including gross-versus-net comparisons and cumulative and rolling-performance evidence. Section 4.2 presents the factor-adjusted performance (alpha) analysis and related inference. Section 4.3 evaluates robustness across subperiods using full-sample selected strategies. Section 4.4 reports predictive diagnostics for regime classification and volatility-signal quality to help interpret the economic results. Finally, Section 4.5 synthesizes the main empirical findings and provides a transition to the discussion.

4.1 Performance Results

4.1.1 Out-of-Sample Net Performance: Regime-Switching ML Strategies

Table 7: Performance Comparison: Signal Power vs. Realizability (1980–2024)

Basket Strategy	Best ML Model	Ann. Ret. (%)	CAGR (%)	Vol. (%)	Sharpe	Sortino	Max DD (%)	DD Date
Panel A: Gross Returns (Signal Power)								
Beta Expansion	LSTM (Rolling 10Y)	16.85	15.88	19.96	0.649	0.933	-47.5	2009-02
Rolling Beta	LSTM (Rolling 10Y)	16.71	15.75	19.86	0.646	0.931	-44.8	2009-02
Kalman Filter Beta	LSTM (Rolling 10Y)	16.58	15.60	19.86	0.639	0.918	-44.9	2009-02
EWMA Beta	LSTM (Rolling 10Y)	16.55	15.57	19.86	0.638	0.913	-45.8	2009-02
Economic Basket	XGBoost	15.93	14.78	20.22	0.596	0.860	-49.0	2009-02
NYSE Benchmark	—	14.71	13.95	17.66	0.613	0.846	-48.4	2009-02
Panel B: Net Returns (Realizability)								
Beta Expansion	LSTM (Rolling 10Y)	16.47	15.45	19.97	0.631	0.906	-47.8	2009-02
Rolling Beta	LSTM (Rolling 10Y)	16.34	15.32	19.87	0.627	0.903	-45.1	2009-02
Kalman Filter Beta	LSTM (Rolling 10Y)	16.20	15.16	19.87	0.620	0.890	-45.2	2009-02
EWMA Beta	LSTM (Rolling 10Y)	16.16	15.13	19.86	0.618	0.885	-46.1	2009-02
Economic Basket	XGBoost	15.54	14.34	20.22	0.577	0.832	-49.3	2009-02
NYSE Benchmark	—	14.71	13.95	17.66	0.613	0.846	-48.4	2009-02

Notes: This table compares the performance of the best-performing ML model for each basket strategy under two conditions: gross returns (Panel A, signal power) and net returns (Panel B, realizability after transaction costs). The sample consists of NYSE/AMEX/NASDAQ stocks with price $\geq$ \$5 and market capitalization $\geq$ 20th percentile (NYSE) from July 1980 to June 2024 (528 months). The Market Benchmark is an equal-weighted portfolio of all eligible stocks with monthly rebalancing (no transaction costs). Panel A measures *signal power* (returns before costs), while Panel B measures *realizability* (returns after Frazzini (2018) transaction costs, applied annually in June). For each basket, the best ML model is selected based on the highest Sharpe ratio among five models: Logistic Regression (L2), Random Forest, XGBoost, LSTM (Expanding Window), and LSTM (Rolling 10Y). *Ann. Ret.* = annualized arithmetic return, *CAGR* = compound annual growth rate, *Vol.* = annualized volatility, *Sharpe* = Sharpe ratio, *Sortino* = Sortino ratio, *Max DD* = maximum drawdown, *DD Date* = date of maximum drawdown.

Table 7 reports out-of-sample performance for the regime-switching allocation across five basket construction methods. Results are presented in two panels to distinguish between *signal power* (gross returns, Panel A) and *economic realizability* (net returns after transaction costs, Panel B).

Panel A: Signal Power. Panel A shows that the dynamic beta-based baskets deliver gross Sharpe ratios between 0.638 and 0.649, compared with 0.596 for the fixed Economic basket and 0.613 for the NYSE Benchmark. The best gross performance is achieved by the Beta-Expansion basket paired with an LSTM (Rolling 10Y) model (Sharpe = 0.649), with annualized return 16.85% and CAGR 15.88%. Across all basket constructions, maximum drawdowns cluster around the Global Financial Crisis trough (Feb. 2009), indicating that improvements primarily reflect higher risk-adjusted performance rather than avoidance of the common crisis drawdown.

Panel B: Realizability. Panel B incorporates transaction costs and therefore reflects implementable performance. Net performance remains close to gross performance, consistent with the fact that holdings-level trading occurs primarily at the annual June rebalancing date and transaction costs scale with realized turnover. The Beta-Expansion basket with LSTM (Rolling 10Y) remains the top performer after costs (net Sharpe = 0.631; annualized return 16.47%; CAGR 15.45%). The remaining dynamic baskets retain net Sharpe ratios between 0.618 and 0.627. The Economic basket remains below the benchmark on a risk-adjusted basis (net Sharpe = 0.577 vs. 0.613), despite a higher net annualized return than the benchmark (15.54% vs. 14.71%).

To contextualize the gross-to-net wedge, Appendix Table 23 reports realized turnover by basket (stock-level rebalancing and inter-regime switching), which maps directly into the transaction-cost drag.

Model patterns. Across both panels, the best-performing specifications are consistently obtained by the LSTM (Rolling 10Y) model for the dynamic beta baskets, whereas XGBoost is selected for the fixed Economic basket. This pattern suggests that capturing regime persistence and state dependence is valuable for the allocation problem; formal inference is deferred to the factor-model alpha analysis in Section 4.2.

Remark. For each basket, the reported “best” ML model is selected based on the highest Sharpe ratio among the candidate models. To mitigate selection concerns, Table 22 in Appendix D.1 reports the complete performance grid across all models (gross and net). In addition, Table 24 in Appendix D.2.1 reports realized transaction costs and their decomposition

(stock-level rebalancing versus inter-regime switching). Table 23 in the Appendix documents the underlying turnover that drives these costs, highlighting that the cost drag is primarily driven by turnover rather than by the cost schedule alone.

Before evaluating the regime-timing strategies, it is useful to characterise the *unconditional* properties of the two allocation sleeves that the timing rule can choose from. Specifically, each basket is split into an *Offensive* and a *Defensive* sleeve, and we report their standalone performance over 1980–2024. Table 8 summarises annualised return moments (annualised return, CAGR, volatility), downside-risk metrics (Sortino, maximum drawdown), and the timing of the worst drawdown. Panel A reports gross returns, while Panel B applies the transaction-cost schedule described in Appendix D.2.1. Importantly, the reported numbers are *averages across ML models* for a given (Basket, Regime) sleeve; this table is therefore not intended to identify a single “best” predictive model, but to document the baseline characteristics of the investable building blocks used in the dynamic allocation stage.

4.1.2 Basket Building Blocks: Offensive vs. Defensive Sleeves (Gross vs. Net)

Table 8: Basket-Level Performance Analysis: Offensive vs Defensive Strategies (1980–2024)

Panel	Basket	Regime	Ann. Ret. (%)	CAGR (%)	Vol. (%)	Sharpe	Sortino	Max DD (%)	DD Date
Panel A: Gross Returns (No Transaction Costs)									
	Economic Basket	Offensive	15.64	14.41	20.36	0.577	0.840	-49.0	2009-02
		Defensive	15.67	14.62	19.68	0.599	0.873	-46.6	2009-02
	Beta Expansion	Offensive	16.08	14.82	20.80	0.587	0.869	-47.5	2009-02
		Defensive	16.11	15.35	18.62	0.657	0.929	-47.0	2009-02
	Rolling Beta	Offensive	16.25	14.94	21.12	0.585	0.877	-44.8	2009-02
		Defensive	15.86	15.10	18.39	0.652	0.912	-49.2	2009-02
	EWMA Beta	Offensive	16.06	14.70	21.22	0.574	0.857	-45.8	2009-02
		Defensive	16.04	15.32	18.32	0.664	0.924	-48.0	2009-02
	Kalman Filter Beta	Offensive	16.20	14.89	21.14	0.583	0.876	-44.9	2009-02
		Defensive	15.85	15.11	18.32	0.654	0.911	-48.0	2009-02
	NYSE Benchmark	—	14.71	13.95	17.66	0.613	0.846	-48.4	2009-02
Panel B: Net Returns (With Transaction Costs)									
	Economic Basket	Offensive	15.24	13.97	20.37	0.558	0.812	-49.3	2009-02
		Defensive	15.33	14.22	19.69	0.581	0.847	-46.8	2009-02
	Beta Expansion	Offensive	15.69	14.38	20.81	0.567	0.840	-47.7	2009-02
		Defensive	15.75	14.94	18.62	0.638	0.903	-47.4	2009-02
	Rolling Beta	Offensive	15.86	14.50	21.13	0.566	0.847	-45.0	2009-02
		Defensive	15.49	14.69	18.39	0.632	0.885	-49.4	2009-02
	EWMA Beta	Offensive	15.65	14.23	21.23	0.554	0.826	-46.1	2009-02
		Defensive	15.68	14.91	18.32	0.645	0.897	-48.3	2009-02
	Kalman Filter Beta	Offensive	15.81	14.44	21.15	0.564	0.846	-45.1	2009-02
		Defensive	15.48	14.69	18.32	0.633	0.885	-48.4	2009-02
	NYSE Benchmark	—	14.71	13.95	17.66	0.613	0.846	-48.4	2009-02

Note: Performance metrics averaged across all ML models within each (Basket, Regime) pair. Transaction costs applied once per year in June following Fama-French convention. Inter-basket costs (High/Low Beta switching) included in strategy-specific TC. Baskets sorted by Sharpe ratio (descending) within each panel.

Table 8 decomposes basket-level performance by allocation sleeve (Offensive vs. Defensive), reporting metrics averaged across ML models. Three empirical regularities stand out. First, across all basket constructions, the Defensive sleeve exhibits materially lower volatility (about 18.3–19.7% versus roughly 20.4–21.2% for Offensive), which translates into higher risk-adjusted performance. Defensive Sharpe ratios range from 0.60 to 0.66 in gross terms and from 0.58 to 0.65 net of transaction costs, whereas Offensive Sharpe ratios are generally lower (about 0.57–0.59 gross and 0.55–0.57 net).

Second, the gross-to-net wedge is economically modest: annualized returns typically decline by roughly 35–45 bps after costs in both sleeves, and Sharpe ratios fall by only about 0.01–0.03. This pattern is consistent with trading frictions being driven primarily by turnover at the annual June reconstitution, with limited additional month-to-month trading at the holdings level.

Third, relative to the NYSE Benchmark, both sleeves earn comparable or higher average returns, but the Defensive sleeve is the only one that consistently matches or exceeds the benchmark Sharpe ratio, indicating that its advantage primarily reflects volatility (and downside-risk) reduction rather than a higher unconditional risk premium. Finally, maximum drawdowns remain concentrated around the Global Financial Crisis trough (Feb. 2009) for all sleeves, highlighting that

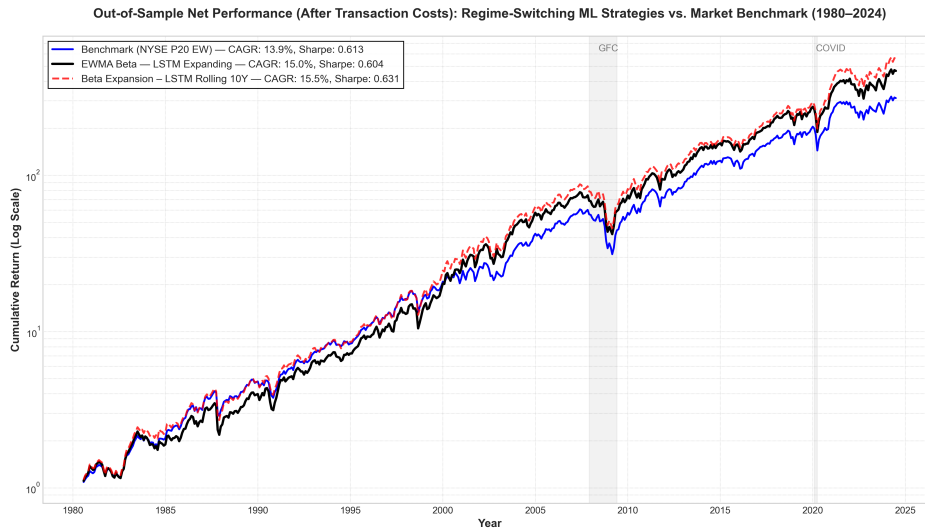
these long-only sleeves mitigate risk but do not constitute crash-hedging strategies.

These unconditional sleeve characteristics motivate the regime-switching exercise: since Offensive and Defensive exposures offer similar long-run growth but different risk profiles, the potential value of predictability lies in allocating between them over time. We therefore turn to the out-of-sample performance of regime-switching strategies that dynamically select the sleeve based on predicted volatility regimes. Table 7 reports performance net of transaction costs and benchmarks it against the market and the corresponding buy-and-hold basket.

4.1.3 Cumulative Net Performance

Summary statistics can mask when performance differences occur. To complement Table 7 and illustrate the long-run economic magnitude of net performance, Figure 4 plots cumulative net wealth (log scale) for the two best-performing regime-switching ML strategies alongside the market benchmark.

Figure 4: Out-of-Sample Net Cumulative Performance (After Transaction Costs): Best Regime-Switching ML Strategies vs. Market Benchmark (1980–2024)



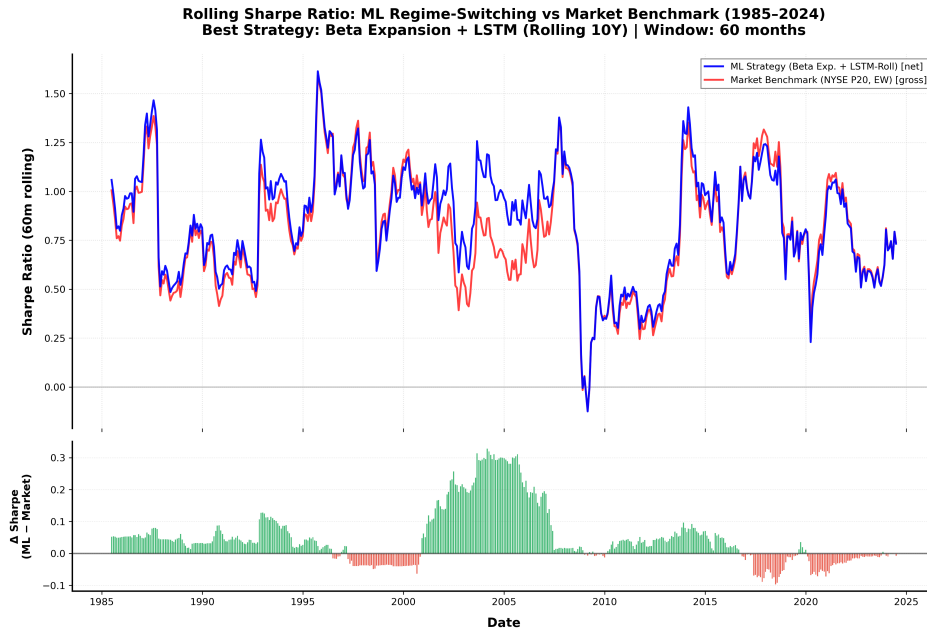
Notes: This figure plots log cumulative wealth (starting at 1) for the NYSE Benchmark and the two best-performing regime-switching ML strategies (EWMA Beta with LSTM expanding window; Beta Expansion with LSTM rolling 10-year window) over July 1980–June 2024. Strategy returns are reported net of transaction costs using the Time-varying institutional cost schedul applied at the annual June rebalancing (including inter-regime switching costs where applicable). The benchmark is shown gross (no transaction costs) for comparability with Table 7. Shaded areas highlight the Global Financial Crisis (2007–2009) and the COVID-19 crash (2020).

Figure 4 plots cumulative *net* performance (after transaction costs) of the two top regime-switching ML strategies against the market benchmark over 1980–2024 on a log scale. The log representation highlights differences in long-run growth rates: both ML strategies compound faster than the benchmark over the full sample, consistent with the higher net CAGRs reported in Table 7. Importantly, the relative ordering is stable across major stress episodes (GFC and COVID), suggesting that the outperformance is not driven by a single period. While the figure is descriptive, it motivates the next step of the analysis: testing whether the performance differential reflects compensation for systematic risk exposures or persistent abnormal returns (alpha) in factor models.

Cumulative wealth curves provide a long-run summary but can obscure shorter-lived episodes of relative outperformance and underperformance. To assess the stability of performance through time, Figure 5 reports 60-month rolling Sharpe ratios for the best-performing ML regime-switching strategy and the market benchmark, together with the rolling Sharpe differential (Δ Sharpe).

4.1.4 Time-Varying Performance: Rolling Sharpe and Relative Outperformance

Figure 5: Rolling Sharpe Ratio and Outperformance: ML vs Market Benchmark



Note: *Panel A* shows the 60-month rolling Sharpe ratio (annualized) for the best-performing ML regime-switching strategy (solid blue line, net of transaction costs) and the market benchmark (red line, NYSE 20th-percentile market-cap breakpoint equal-weighted portfolio, gross returns). *Panel B* displays the difference in Sharpe ratios ($\Delta \text{Sharpe} = \text{ML} - \text{Market}$), with green bars indicating periods where the ML strategy outperforms (positive Δ) and red bars indicating underperformance (negative Δ). The ML strategy is selected based on the maximum net Sharpe ratio over the full sample (July 1980–June 2024). Rolling Sharpe ratios are computed using a 60-month backward-looking window, with the first observation in January 1985. The ML strategy outperforms the market over extended subperiods, although episodes of underperformance also occur. The benchmark is shown gross (no transaction costs), implying that ΔSharpe is conservative with respect to net implementability.

Figure 5 shows that the ML strategy’s risk-adjusted performance advantage is time-varying: rolling Sharpe differentials fluctuate and occasionally turn negative, as expected in a realistic time-series setting. However, positive ΔSharpe dominates over large parts of the sample, with several multi-year stretches of sustained outperformance. While the figure is descriptive, it motivates the factor-model analysis below, which evaluates whether the observed outperformance is explained by systematic exposures or reflects residual alpha.

The performance results above are economic and descriptive. The key question for interpretation is whether the observed return differentials are simply compensation for systematic risk exposures (possibly time-varying) or whether they represent residual abnormal performance. We therefore turn to standard asset-pricing regressions to decompose returns into factor exposures and alpha.

4.2 Factor-Adjusted Performance (Alpha Analysis)

4.2.1 Factor Alphas: Buy-and-Hold Sleeves vs. Regime-Switching ML Strategies

We next assess whether the return differentials documented above reflect compensation for systematic risk exposures or abnormal performance. To this end, we estimate time-series regressions of monthly *net* excess returns on standard factor sets. Table 9 reports annualized alphas and Newey–West (HAC, 12 lags) t -statistics under the Fama–French six-factor model (FF6) and the q-factor specification (Q5). Panel A summarizes the buy-and-hold baskets (Offensive/Defensive), while Panel B reports the best-performing ML strategy within each basket.

Table 9: Factor-Adjusted Performance of Buy&Hold Baskets and ML Strategies (Net)

Basket	Strategy	FF6			Q5			
		α (%)	t	R^2	α (%)	t	R^2	
Panel A: Buy&Hold Baskets (Net)								
Economic	(Offensive)	1.26		0.96	0.939	2.11	1.31	0.904
	(Defensive)	2.14	*	1.68	0.954	2.22	1.36	0.936
Beta Exp.	(Offensive)	2.82	*	1.94	0.951	3.00	1.59	0.934
	(Defensive)	1.28		1.17	0.945	1.80	1.22	0.911
Rolling β	(Offensive)	2.91	*	1.80	0.945	3.25	1.57	0.929
	(Defensive)	1.08		1.03	0.939	1.37	0.96	0.903
EWMA β	(Offensive)	2.62	*	1.70	0.945	2.98	1.51	0.932
	(Defensive)	1.33		1.19	0.937	1.64	1.12	0.900
Kalman β	(Offensive)	2.78	*	1.84	0.945	3.03	1.56	0.931
	(Defensive)	1.08		0.96	0.938	1.49	0.99	0.901
Panel B: Best ML Strategies by Basket (Net)								
Economic	XGB	1.73		1.31	0.941	2.14	1.32	0.908
Beta Exp.	RF	2.77	*	1.94	0.948	2.77	1.49	0.932
Rolling β	RF	3.14	**	2.02	0.940	2.99	1.49	0.926
EWMA β	RF	2.80	*	1.83	0.938	2.58	1.31	0.927
Kalman β	RF	2.82	*	1.91	0.940	2.59	1.35	0.927

Notes: This table reports risk-adjusted performance metrics for buy-and-hold baskets (Panel A) and the best-performing ML strategy per basket (Panel B). All returns are net of transaction costs. Alphas (α) are annualized and expressed in percentage points. t -statistics use Newey–West HAC standard errors with 12 lags. R^2 measures the explanatory power of the factor models. FF6 includes market, size, value, profitability, investment, and momentum factors. Q5 includes market, size, investment, profitability, and expected growth factors. Sample period: July 1980–June 2024. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

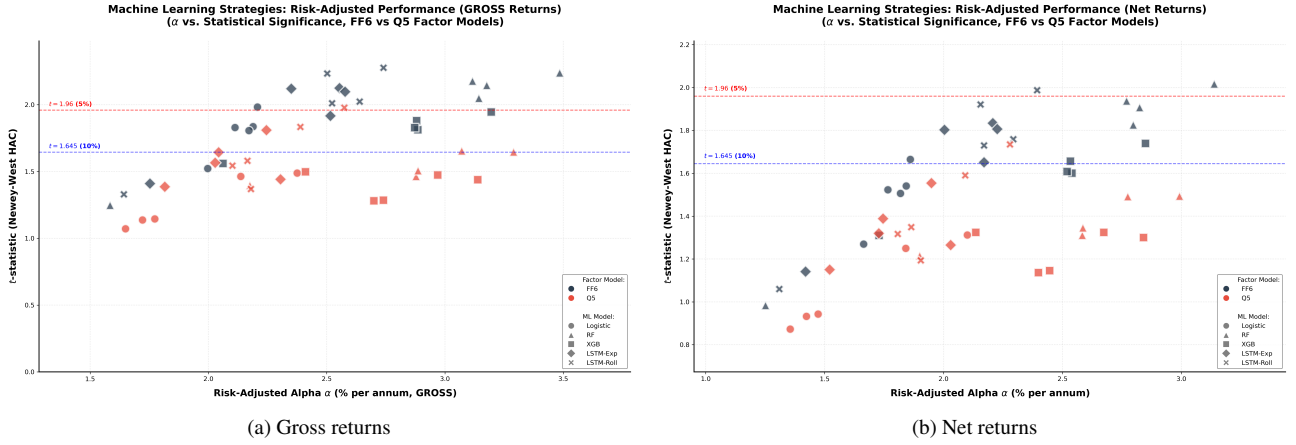
Two patterns emerge. First, the buy-and-hold sleeves in Panel A deliver positive factor-model alphas, but statistical significance is generally modest: most Newey–West t -statistics cluster around 1–2, implying that abnormal performance is often only marginally significant once common risk factors are controlled for. Second, the best ML strategies in Panel B exhibit comparable or somewhat higher alphas in several baskets—most notably for the Rolling- β basket under FF6 (with $t \approx 2$)—suggesting that regime-based timing can improve risk-adjusted performance beyond passive exposure in specific cases. However, this improvement is not uniform across baskets (e.g., the Economic basket shows weaker statistical evidence), and significance attenuates under the Q5 model for both buy-and-hold and ML strategies. Overall, inference is factor-model dependent, motivating incremental-alpha tests relative to fixed buy-and-hold benchmarks and additional robustness checks reported in the Appendix.

For completeness, Appendix D.3 (Table 25) reports the full set of factor regressions for all buy-and-hold sleeves and all 25 ML strategies under FF5/FF6/FF7 and Q5.

4.2.2 Alpha Magnitudes vs. Statistical Significance Across ML Specifications

Table 9 focuses on a small set of representative strategies (buy-and-hold sleeves and the best ML strategy per basket). To separate *signal power* from *implementability*, Figure 6 contrasts factor-model alphas estimated on gross returns (Panel a) with the corresponding alphas estimated on net returns after transaction costs (Panel b).

Figure 6: Machine Learning Strategies: Risk-Adjusted Alpha vs. Statistical Significance (FF6 vs. Q5, Gross vs. Net)



Notes: This figure plots, for each of the 25 machine-learning strategies (five baskets $\times$ five ML models), the annualized factor-model alpha (α , in percent per annum) against its Newey–West HAC t -statistic (12 lags). Points are shown under two alternative risk-adjustment models: the Fama–French six-factor model (FF6) and the Hou–Xue–Zhang q -factor model (Q5). Panel (a) reports alphas estimated on gross returns; Panel (b) reports alphas estimated on returns net of transaction costs. Horizontal dashed lines mark conventional significance thresholds ($|t| = 1.96$ for 5% and $|t| = 1.645$ for 10%). Values closer to the upper-right corner indicate larger and more statistically significant abnormal performance. Sample period: July 1980–June 2024.

Figure 6 summarizes cross-strategy dispersion in risk-adjusted performance by plotting annualized alphas against Newey–West t -statistics for each of the 25 ML specifications. Two insights emerge. First, moving from gross (Panel a) to net returns (Panel b) shifts the cloud of points primarily leftward (lower α), with a more modest reduction in statistical support. This pattern indicates that transaction costs mainly erode the economic magnitude of abnormal performance rather than mechanically eliminating all predictability, consistent with trading being concentrated in discrete rebalancing decisions (and, where applicable, additional sleeve-switching trades). Second, inference remains factor-model dependent in both panels: relative to FF6, the Q5 specification generally delivers weaker t -statistics for a given strategy, suggesting that part of the apparent abnormal performance can be absorbed by alternative systematic exposures. Overall, the gross-versus-net comparison highlights that only a subset of specifications retains economically meaningful and statistically supported alphas after costs, motivating the incremental benchmark tests reported next.

To sharpen the interpretation, we complement Table 9 with *incremental-alpha* tests that benchmark each ML strategy against transparent buy-and-hold allocations. Table 10 reports $\Delta\alpha$ relative to a fixed 50/50 benchmark (50% Offensive + 50% Defensive), which is ex-ante neutral and not informed by future performance. The main results are economically concentrated in a subset of specifications, while the cross-model mean and median $\Delta\alpha$ remain modest, underscoring that the gains are not ubiquitous across ML choices. Appendix Table 26 extends this analysis to the full set of 25 strategies under FF6 and Q5 and reports FDR-adjusted q -values to account for multiple testing. Further robustness to alternative benchmarks (Offensive-only, Defensive-only), alternative factor specifications, and subsamples is reported in Appendix Tables 27, 28, and 29. Finally, Appendix Tables 24 and 31 document that implementability is primarily driven by turnover-related costs rather than the cost schedule alone.

4.2.3 Incremental Alpha Relative to a 50/50 Buy-and-Hold Benchmark

To isolate the contribution of regime timing relative to a transparent passive allocation, we estimate *incremental alphas* on net return differentials with respect to an ex-ante 50/50 benchmark (50% Offensive + 50% Defensive). Specifically, we run factor regressions on $r_{ML,t}^{net} - r_{50/50,t}^{net}$, where both the ML strategy and the benchmark returns are computed net of transaction costs using the same round-trip cost convention and time-varying cost schedule. Table 10 reports $\Delta\alpha$ for the best model in each basket, along with summary statistics across all 25 ML specifications and a simple benchmark-robustness panel.

Table 10: Incremental Alpha: ML vs Buy&Hold 50/50 Benchmark

Basket	Best Model	FF6		Q5	
		$\Delta\alpha$ (%)	t	$\Delta\alpha$ (%)	t
Economic	XGB	0.03	0.12	-0.03	-0.10
Beta Exp.	RF	0.72	1.58	0.37	0.71
Rolling β	RF	1.14	*	0.68	0.90
EWMA β	RF	0.82	1.43	0.27	0.38
Kalman β	RF	0.89	1.61	0.33	0.46
Average (all models)	–	0.24	0.35	-0.19	-0.28
Median (all models)	–	0.21	0.30	-0.22	-0.46

Robustness: Alternative Benchmarks (FF6 only)			
Benchmark	Mean $\Delta\alpha$ (%)	% $\Delta\alpha > 0$ (FF6)	# Significant (FF6)
50/50 (main)	0.24	64%	0/25
Offensive only	-0.31	28%	0/25
Defensive only	0.79	80%	1/25

Notes: This table reports incremental alphas ($\Delta\alpha$, annualized in % p.a.) obtained from factor regressions on the *net return differential* between each ML strategy and a passive benchmark, $r_{ML,t}^{net} - r_{bench,t}^{net}$. The main panel reports results versus an ex-ante 50/50 benchmark (50% Buy&Hold Offensive + 50% Buy&Hold Defensive), and shows the best-performing ML model per basket (selected based on the highest net Sharpe ratio in the performance analysis). Summary rows report the mean and median $\Delta\alpha$ across the full set of 25 ML strategies (5 baskets $\times$ 5 models). The robustness panel compares incremental alphas across alternative benchmarks (FF6 only): the main 50/50 benchmark, an Offensive-only benchmark, and a Defensive-only benchmark. *Significant* counts the number of strategies with $|t| > 1.96$ (5% level, non-FDR-adjusted). All strategy and benchmark returns are computed net of transaction costs using the same time-varying cost schedule and a round-trip implementation (buy + sell); costs are applied proportionally to realized turnover under the annual June rebalancing convention, and include inter-basket switching costs where applicable. t -statistics use Newey–West HAC standard errors with 12 lags on monthly data. Sample period: July 1980–June 2024. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Overall, incremental performance relative to a passive 50/50 allocation is economically modest and not pervasive across ML choices. Under FF6, only the strongest specification delivers marginal statistical support (10% level), while significance generally vanishes under Q5, highlighting model dependence. Results are also sensitive to the benchmark used for differencing, as shown in the robustness panel. Appendix Table 26 reports the full grid of 25 strategies under FF6 and Q5 and includes FDR-adjusted q -values to account for multiple testing. Robustness to alternative benchmarks and factor specifications is reported in Appendix Tables 27 and 28, and subperiod evidence is summarized in Appendix Table 29. Finally, implementability is assessed via realized cost and break-even analyses in Appendix Tables 24 and 31.

Since the incremental-alpha evidence is not uniform across models and appears sensitive to both the risk-adjustment specification and the benchmark choice, a natural next question is whether the results are stable through time or concentrated in specific market eras. We therefore conduct a subperiod analysis that evaluates full-sample selected strategies without re-optimization.

4.3 Subperiod Analysis

4.3.1 Stability Across Time: Subperiod Performance of Full-Sample Selected Strategies

To assess whether full-sample outperformance is concentrated in particular eras, we re-evaluate the full-sample selected (Basket, Model) pairs within three subperiods *without any re-optimization*. Table 11 reports subperiod Sharpe ratios computed from monthly *net* returns and incremental alphas relative to the 50/50 buy-and-hold benchmark under FF6.

Table 11: Subperiod Performance of Full-Sample Selected ML Strategies vs. 50/50 Benchmark (FF6)

Basket	Model	P1: 1980–1999			P2: 2000–2009		P3: 2010–2024	
		Sharpe	$\Delta\alpha$		Sharpe	$\Delta\alpha$	Sharpe	$\Delta\alpha$
Economic	XGB	0.90	0.63		0.61	-0.35	0.74	-0.31
Beta Exp.	LSTM-Roll	0.94	1.12	*	0.67	-0.27	0.80	0.58
Rolling β	LSTM-Roll	0.94	1.73	**	0.70	0.10	0.77	0.33
EWMA β	LSTM-Roll	0.93	1.31	*	0.71	0.23	0.75	-0.26
Kalman β	LSTM-Roll	0.93	1.36	**	0.70	0.26	0.76	-0.07

Notes: The set of strategies (Basket, Model) is selected once using the full sample (July 1980–June 2024) based on maximum net Sharpe ratio, and then evaluated within each subperiod without re-optimization. $\Delta\alpha$ denotes the incremental alpha (in % per annum) of the ML strategy relative to the 50/50 benchmark (50% Buy&Hold Offensive + 50% Buy&Hold Defensive) under the Fama–French 6-factor (FF6) model. Subperiods are: P1 (1980–1999), P2 (2000–2009), P3 (2010–2024). Sharpe ratios are computed from monthly net returns (annualized). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 11 highlights two regularities. First, incremental performance is strongest in the early sample (1980–1999), where the dynamic beta baskets exhibit economically large and, in several cases, statistically significant $\Delta\alpha$ (ranging from about 1.3% to 1.7% p.a., with the largest effect for Rolling β). Second, the incremental edge weakens materially in the 2000–2009 subperiod and is generally small in 2010–2024, suggesting that the timing component is episodic rather than uniform. This pattern is consistent with the strategy adding most value in environments with more pronounced regime variation.

While Table 11 documents that the time pattern of incremental performance is concentrated in the early sample under FF6, alpha inference may depend on the chosen risk-adjustment model. We therefore repeat the same subperiod exercise under alternative factor specifications.

4.3.2 Robustness Across Factor Models: Subperiod Incremental Alphas Under FF5/FF6/FF7 and Q5

Table 12 reports subperiod $\Delta\alpha$ estimates under FF5, FF6, and FF7 (Panel A) and under Q5 (Panel B), using the same 50/50 benchmark and the same net-of-cost return construction as in the main subperiod analysis.

Table 12: Subperiod Robustness: All Factor Models (FF5, FF6, FF7, Q5)

Panel A: Fama–French Factor Models (FF5, FF6, FF7)													
Basket	Model	P1 (1980–1999)						P2 (2000–2009)			P3 (2010–2024)		
		FF5		FF6		FF7		FF5	FF6	FF7	FF5	FF6	FF7
Economic	XGB	0.36		0.63		0.51		-0.36	-0.35	-0.42	-0.30	-0.31	-0.26
Beta Exp.	LSTM-Roll	1.00	**	1.12	*	1.35	**	-0.24	-0.27	-0.31	0.60	0.58	0.81
Rolling β	LSTM-Roll	1.50	**	1.73	**	2.05	***	0.13	0.10	0.21	0.27	0.33	0.26
EWMA β	LSTM-Roll	1.16	*	1.31	*	1.59	**	0.24	0.23	0.36	-0.30	-0.26	-0.27
Kalman β	LSTM-Roll	1.28	**	1.36	**	1.61	**	0.28	0.26	0.39	-0.08	-0.07	-0.06
Panel B: q -Factor Model (Q5)													
Basket	Model	P1 (1980–1999)						P2 (2000–2009)			P3 (2010–2024)		
		Q5						Q5			Q5		
Economic	XGB	0.77						0.26			-0.26		
Beta Exp.	LSTM-Roll	0.72						-1.60			0.77		
Rolling β	LSTM-Roll	1.14						-1.53			0.27		
EWMA β	LSTM-Roll	0.71						-1.27			-0.33		
Kalman β	LSTM-Roll	0.80						-1.26			-0.12		

Notes: Incremental alphas ($\Delta\alpha$, % p.a.) vs. 50/50 benchmark under all factor models. Panel A: FF5, FF6, FF7. Panel B: Q5. Subperiods: P1 (1980–1999), P2 (2000–2009), P3 (2010–2024). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 12 confirms that the time concentration of incremental performance is robust within the Fama–French family: $\Delta\alpha$ is economically largest in P1 and generally attenuates in P2–P3 across FF5/FF6/FF7. At the same time, inference is strongly factor-model dependent. In particular, the Q5 specification delivers markedly weaker incremental performance in P2 (often turning negative for the dynamic beta baskets) and provides little support for a persistent edge in the later subperiod. Overall, the subperiod evidence suggests that regime-timing benefits are state-dependent and that conclusions about “alpha” hinge on the chosen risk-adjustment benchmark.

Appendix Table 30 extends this subperiod analysis to the full grid of 25 strategies across factor models (FF5/FF6/FF7 and Q5).

Having established the economic performance of the regime-switching strategies, we next examine whether the underlying volatility-regime signal is predictively meaningful out of sample. This diagnostic section evaluates the classification quality of the regime forecasts and the behavior of the EWMA volatility proxy that drives the allocation rule.

4.4 Predictive Diagnostics: Regime Classification and Volatility Signal Quality

The performance results above rely on a volatility-based regime signal and an annual (June) prediction step. To interpret the economic gains, we therefore report (i) out-of-sample regime-classification metrics at the portfolio formation date, (ii) the time-series behaviour of the EWMA volatility signal and the dynamic regime threshold, and (iii) forecast calibration statistics that quantify how closely ex-ante volatility maps into ex-post realized volatility.

4.4.1 Out-of-Sample Regime-Classification Performance (Annual June Decisions)

Table 13 summarizes the predictive performance of the competing ML classifiers at the annual portfolio-formation date (June), with one out-of-sample regime forecast per year for the subsequent July–June holding period. This evaluation is aligned with the trading design and therefore reflects the information set available to an investor at implementation time.

Table 13: Machine Learning Models: Regime Classification Performance (Annual June Decisions)

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Bal. Acc.	F1 Macro	Brier	Log Loss
Logistic	0.7273	0.6667	0.5000	0.5714	0.8393	0.6786	0.6857	0.1721	0.5192
Random Forest	0.7045	0.6667	0.3750	0.4800	0.7165	0.6339	0.6368	0.1967	0.5754
XGBoost	0.6591	0.5385	0.4375	0.4828	0.7321	0.6116	0.6143	0.2256	0.7140
LSTM (Exp.)	0.7273	0.7000	0.4375	0.5385	0.8192	0.6652	0.6725	0.1679	0.5006
LSTM (Roll.)	0.7273	0.6250	0.6250	0.6250	0.7076	0.7054	0.7054	0.2116	0.6492

Notes: Classification metrics are evaluated at the annual portfolio-formation date (June), with one out-of-sample regime forecast per year for the subsequent July–June holding period, consistent with the Fama–French convention. Walk-forward training starts with an initial training window (1969–1979), and the out-of-sample evaluation spans 1980–2023 ($N = 44$ annual decisions). Lower values indicate better calibration for Brier score and log loss; higher values are better for all other metrics. Bold values indicate the best model by each metric.

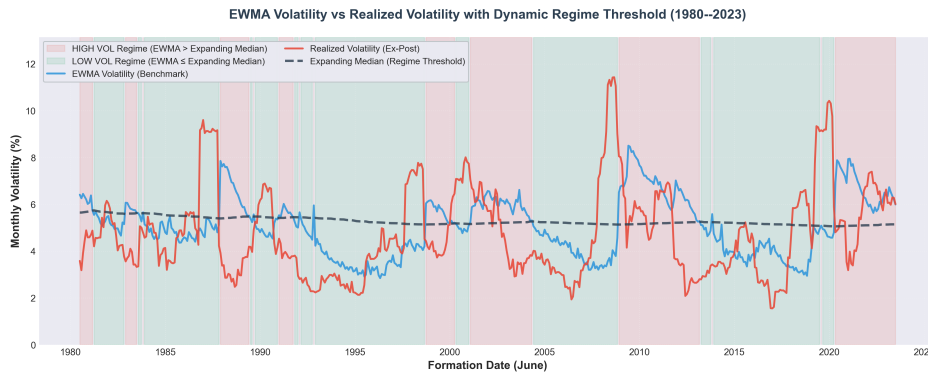
Table 13 suggests that no single model dominates across all criteria. Logistic regression achieves the highest ROC-AUC, indicating strong ranking ability, while LSTM (Rolling) attains the best balance of precision and recall (highest F1-type metrics). Calibration-oriented losses (Brier score and log loss) are lowest for LSTM (Exp.), suggesting slightly better probabilistic forecasting. Overall, the differences are economically modest, which is consistent with the difficulty of forecasting volatility regimes from a small number of annual out-of-sample decisions.

Since regime decisions are ultimately driven by the underlying volatility proxy and its threshold rule, we next visualize how the EWMA signal evolves through time relative to realized volatility and how often it classifies periods as high- versus low-volatility.

4.4.2 EWMA Volatility Signal and Dynamic Regime Threshold

Figure 7 plots the EWMA volatility signal formed at each June date together with the subsequent realized volatility over the next 12 months. The dashed line shows the expanding-median threshold used to separate high- and low-volatility regimes, and the shaded background indicates the implied regime classification.

Figure 7: EWMA Volatility vs Realized Volatility with Dynamic Regime Threshold



Note: EWMA volatility ($\lambda = 0.94$, 60-month window, MA12 smoothing) computed on NYSE Benchmark returns. Formation: June year t (uses data up to May t). Realized volatility measured over next 12 months (July $t \rightarrow$ June $t + 1$). Regime threshold = expanding median of EWMA (computed on data up to May t applied to prevent lookahead). Sample: 44 annual formations (June 1980–June 2023).

The regime-classification exercise is intentionally aligned with the trading design and is therefore evaluated only at the annual June formation dates, yielding $N = 44$ out-of-sample decisions. By contrast, to assess the informational content of the EWMA volatility proxy itself, we evaluate forecast accuracy using *monthly forecast origins* from June 1980 through June 2023 ($N = 517$ months). Each forecast, formed at month-end t using market returns through $t - 1$, is paired with realized volatility measured over the subsequent 12-month window $[t + 1, t + 12]$. The final evaluable forecast (June 2023) uses information through May 2023 and evaluates realized volatility through June 2024, ensuring temporal alignment with the backtest while maximizing the out-of-sample evaluation window.

Figure 7 highlights that the EWMA proxy is smooth and persistent, while realized volatility exhibits sharper spikes around major stress episodes. The expanding-median threshold is deliberately slow-moving, which stabilizes regime assignment and limits excessive switching. As a result, the classification captures broad high-volatility episodes rather than attempting to track month-to-month fluctuations in realized volatility—a design choice consistent with annual portfolio formation.

While the figure is informative about timing and persistence, it remains descriptive. We therefore report simple calibration statistics to quantify the accuracy of the volatility proxy as a forecast of subsequent realized volatility.

4.4.3 Calibration of the Volatility Forecast

Table 14 reports forecast-quality statistics for the EWMA volatility proxy, comparing the ex-ante signal to ex-post realized volatility over the next 12 months. The goal is not to optimize point forecasts, but to assess whether the proxy contains reliable information for regime classification.

Table 14: EWMA Volatility Forecast Calibration Quality (1980–2023)

Correlation	Regime Acc. (%)	Bias (%)	MAE (%)	RMSE (%)	N
0.083	54.4	-0.327	1.78	2.25	517

Notes: *Correlation* is the Pearson correlation between the EWMA volatility forecast $\hat{\sigma}_t$ (formed at forecast date t using information available up to t) and ex-post realized volatility σ_t^{real} computed over the subsequent 12 months ($t+1$ to $t+12$; i.e., July t to June $t+1$ when t is a June formation date). *Regime Acc.* is the fraction of observations for which the forecast and realized volatilities fall on the same side of their respective sample medians, i.e., $\mathbb{P}\{\hat{\sigma}_t > \text{median}(\hat{\sigma})\} = \mathbb{P}\{\sigma_t^{\text{real}} > \text{median}(\sigma^{\text{real}})\}$. *Bias* is the mean forecast error $\mathbb{E}[\sigma_t^{\text{real}} - \hat{\sigma}_t]$; MAE and RMSE are computed on the same forecast errors. EWMA uses $\lambda = 0.94$ with a 60-month rolling estimation window and MA(12) smoothing, applied to NYSE Benchmark returns. *Sample:* $N = 517$ monthly forecast dates (1980m06–2023m06). Realized volatility uses returns through 2024m06 to form the $t+1$ to $t+12$ windows.

Table 14 indicates limited accuracy for forecasting volatility levels: the correlation between the EWMA signal and subsequent realized volatility is small. However, the proxy still provides modest directional information for coarse regime classification (Regime Acc. = 54.4%), which is the relevant object for the allocation rule. In other words, even a noisy level forecast can be useful when the decision problem is binary (high vs. low volatility) and the threshold is designed to be slow-moving.

4.5 Summary of Findings

This section provides out-of-sample evidence on the economic value of machine-learning-based regime switching for adaptive factor allocation. Overall, regime-switching strategies built on dynamic beta baskets deliver economically meaningful performance improvements relative to transparent benchmarks, and these gains remain largely intact after transaction costs, consistent with trading concentrated at the annual June rebalancing and turnover-scaled cost drag. Factor-model regressions indicate that alphas are generally positive, but statistical support is modest and sensitive to the risk-adjustment benchmark (FF models vs. Q5), with evidence concentrated in a subset of specifications. Incremental-alpha tests relative to a passive 50/50 buy-and-hold allocation confirm that the timing component is not pervasive across ML choices and is benchmark-dependent. Subperiod results further suggest that the incremental edge is state-dependent, being strongest in the early sample and attenuating thereafter. Finally, predictive diagnostics show that while the EWMA volatility proxy has limited accuracy for forecasting volatility levels, it contains modest but useful information for coarse high/low volatility regime classification, which is the relevant object for the allocation rule.

5 Discussion

Our results support a deliberately nuanced interpretation of machine-learning-based regime timing for adaptive factor allocation. On the one hand, several regime-switching configurations deliver higher net Sharpe ratios and higher long-run compounding than transparent benchmarks (Table 7 and Figure 4), and the gross-to-net wedge remains economically modest under our annual June implementation and conservative round-trip cost schedule. On the other hand, the *incremental* contribution of the learning step is neither pervasive nor uniformly statistically supported once we benchmark against simple buy-and-hold alternatives and apply standard risk adjustments. In particular, incremental-alpha tests relative to an ex-ante 50/50 benchmark yield modest $\Delta\alpha$ estimates and limited statistical significance, with inference attenuating under the Q5 specification and after multiple-testing adjustments (Tables 10 and 26). This dispersion is informative: it suggests that “ML timing” is not a generic premium that appears automatically once a classifier is introduced, but depends on (i) how characteristics are aggregated into tradable sleeves and (ii) whether the learned mapping generates economically useful regime decisions at the strategy’s decision frequency.

A central mechanism implied by the full set of results is that the economic value, when present, arises primarily from *coarse regime separation* rather than from accurate volatility *level* forecasting. The volatility-proxy calibration exercise confirms that the EWMA signal has limited ability to forecast realized volatility levels over the subsequent year (Table 14), yet the regime decision problem is binary (high vs. low volatility), and even modest directional information can be useful when combined with a slow-moving threshold and a low-frequency re-optimization design. Put differently, weak point-forecast diagnostics do not mechanically invalidate an allocation rule that only requires stable regime classification, but they do constrain the interpretation: performance should not be attributed to precise volatility forecasting, but to whether the regime rule separates states with meaningfully different sleeve payoffs.

The results also emphasize that turnover and transaction costs are first-order determinants of realizability, but, under our design, they do not dominate the strategy’s economics. Because the main trading activity is concentrated at the annual

June reconstitution, net performance remains close to gross performance for most basket-model combinations (Tables 7 and 22). This implementability is reinforced by the turnover and realized-cost decompositions (Tables 23 and 24), which show that the cost drag is primarily linked to stock-level rebalancing and only secondarily to inter-sleeve switching. At the same time, the break-even analysis clarifies that robustness is benchmark-dependent: headroom is sizable versus the market benchmark for many strategies, but becomes materially smaller when the relevant comparison is a diversified long-only alternative such as the 50/50 benchmark (Table 31). This pattern aligns with the incremental-alpha evidence: much of the apparent performance advantage reflects properties of the underlying sleeves and basket construction, whereas the incremental timing edge is more episodic and less universal.

Subperiod evidence further qualifies the interpretation. When we hold the (Basket, Model) selection fixed and re-evaluate performance without re-optimization, incremental performance is economically strongest in the early sample (1980–1999) and attenuates thereafter (Tables 11 and 12). This suggests that timing benefits are state-dependent and may be linked to periods with more pronounced regime variation, different market microstructure, or a stronger link between volatility states and cross-sectional factor payoffs. Importantly, factor-model dependence remains substantial in subperiods as well: within the Fama–French family, $\Delta\alpha$ patterns are broadly consistent, while the Q5 model absorbs a larger share of apparent abnormal performance in some eras, occasionally reversing conclusions. Overall, the evidence is best read as mapping *where* regime learning is more likely to add incremental value, rather than as establishing a time-invariant “ML timing alpha.”

Several limitations are worth highlighting. First, the regime-classification evaluation at the trading decision frequency is based on a small number of annual out-of-sample decisions (Table 13), which naturally limits statistical power and increases the scope for sampling variation. Second, our regime label is intentionally transparent (EWMA volatility with a median threshold), but it may omit dimensions of market state that matter for factor payoffs; richer regime definitions could change both the economic and diagnostic results. Third, the model-selection step (choosing “best” specifications) is explicitly documented and complemented with full grids and multiple-testing adjustments, yet the presence of many specifications means that isolated maxima should be interpreted cautiously and in the context of cross-specification dispersion. Finally, our transaction-cost schedule is conservative and turnover-scaled, but it remains stylized; cost heterogeneity across stocks, liquidity states, and trade size could matter for institutional implementation.

Taken together, the evidence supports cautious optimism with clear boundaries. Machine learning can add incremental value for regime-based factor allocation in some configurations, especially when combined with basket constructions that create economically distinct sleeves and when evaluated net of realistic costs. However, incremental timing value is not automatic, is sensitive to risk-adjustment benchmarks, and appears episodic across time. Practically, the most defensible approach is therefore to prioritize transparent trading rules, strict timing alignment, net-of-cost evaluation, and reporting that emphasizes distributions across specifications rather than isolated best-case outcomes.

6 Conclusion

This paper evaluates whether machine-learning-based regime detection can improve adaptive allocation between Offensive and Defensive factor sleeves in a transparent, investable long-only setting. We construct 25 characteristic-sorted signal portfolios using strict point-in-time accounting data and investability filters, aggregate them into tradable sleeves under five basket-construction schemes, and implement an annual walk-forward protocol in which a supervised classifier produces a single regime forecast at each June formation date for the subsequent July–June holding period. All components—signals, baskets, features, labels, training windows, and evaluation periods—are aligned under an explicit no-lookahead timing convention to ensure that performance differences reflect the incremental contribution of the learning step rather than calendar misalignment.

Across the resulting grid of basket constructions and model classes, the evidence supports a nuanced conclusion. Regime-switching strategies built on dynamic beta baskets deliver higher net Sharpe ratios and higher long-run compounding than the market benchmark, and these gains remain largely intact after transaction costs under our annual June implementation. However, the *incremental* contribution of machine learning is heterogeneous and benchmark-dependent. When performance is evaluated relative to a transparent diversified long-only alternative (the ex-ante 50/50 benchmark), incremental alphas are economically modest on average and rarely statistically significant once standard factor-model adjustments and multiple-testing considerations are taken into account. Inference is also factor-model dependent: alphas and $\Delta\alpha$ estimates are generally stronger under the Fama–French family and weaker under the Q5 specification.

A practical implication is that the relevant economic mechanism is primarily *coarse regime separation* rather than accurate volatility level forecasting. Our diagnostics confirm that the EWMA volatility proxy has limited accuracy for forecasting realized volatility levels over the subsequent year, yet it can still provide modest information for binary high/low regime classification, which is the object the allocation rule requires. The results therefore suggest that regime learning should be judged by its ability to generate implementable, low-frequency allocation decisions that survive costs and outperform transparent buy-and-hold benchmarks, not by its ability to deliver high-precision volatility forecasts.

The subperiod analysis further indicates that timing benefits are state-dependent. Full-sample selected strategies exhibit

stronger incremental performance in the early sample and attenuated effects in later periods, consistent with the idea that the economic value of timing depends on the strength of regime variation and on the stability of the relationship between volatility states and sleeve payoffs. This time concentration reinforces the importance of reporting out-of-sample performance through time and interpreting “ML timing” as episodic rather than as a uniform premium.

Several limitations qualify the interpretation. The annual decision design yields a small number of out-of-sample regime decisions, constraining statistical power. The regime label is intentionally transparent but may omit other dimensions of market state relevant for factor premia. The large specification grid implies that isolated best-case outcomes should be viewed in the context of dispersion and supported by robustness checks. Finally, while our transaction-cost schedule is conservative and turnover-scaled, richer cost models could refine conclusions about implementability for different investor types.

Overall, the paper’s contribution is methodological and empirical: it provides a fully documented evaluation framework that isolates the incremental role of machine learning in regime-based factor allocation under realistic timing and implementation constraints. The results show that ML can add value in some configurations and eras, but that the effect is neither uniform nor automatic, and must be assessed relative to simple benchmarks and net of costs. Future work could explore alternative regime definitions, decision frequencies, and liquidity-sensitive cost models, as well as the stability of regime-payoff links across international markets and asset classes.

Main takeaways.

- A strict, fully aligned timing protocol is essential: small calendar misalignments in labels/features/formation dates can materially bias apparent “ML timing” gains.
- Net performance improvements versus the market benchmark can be economically non-trivial and remain largely intact after turnover-scaled transaction costs under annual June implementation.
- The incremental contribution of machine learning is heterogeneous and benchmark-dependent: $\Delta\alpha$ relative to a diversified 50/50 buy-and-hold benchmark is modest on average and seldom statistically strong after risk adjustment and multiple-testing considerations.
- The economic mechanism is primarily *regime separation* (high/low classification) rather than precise volatility level forecasting; weak volatility point-forecast calibration does not preclude timing value in a binary decision problem, but it constrains interpretation.
- Timing benefits appear episodic across time (stronger in earlier decades, weaker later), reinforcing the need for subperiod evaluation and cautious generalization.

A Data and Signal Construction

A.1 Fundamental Signal Definitions

Table 15: Signal Definitions and Data Sources

Category	Signal Name	Formula	Zoo ID
<i>Panel A: Profitability</i>			
	Gross Profitability	$\frac{SALE - COGS}{AT}$	#88
	Operating Profitability	$\frac{REVT - COGS - XSGA}{AT}$	#95
	Return on Assets	$\frac{NI}{AT}$	#82
	Δ Asset Turnover	$\Delta \left(\frac{SALE}{AT} \right)$	#72
	Δ Profit Margin	$\Delta \left(\frac{NI}{SALE} \right)$	#74
<i>Panel B: Investment</i>			
	Return on Invested Capital	$\frac{NI - DVC}{CEQ + DLTt}$	#70
	Asset Growth	$\frac{\Delta AT}{AT_{t-1}}$	#92
	Investment (CapEx)	$\frac{CAPX}{AT}$	#53
	CapEx + Inventory Growth	$\frac{\Delta PPEGT + \Delta INVT}{AT}$	#81
	Net Share Issuance	$\frac{\Delta CSHO}{CSHO_{t-1}}$	#73
	CapEx Growth	$\frac{\Delta CAPX}{AT}$	#63
<i>Panel C: Accruals & Quality</i>			
	Sales Growth	$\frac{\Delta SALE}{SALE_{t-1}}$	#24
	Working Capital Accruals	$\frac{\Delta WC}{AT}$	#28
	Sales-Receivables Gap	$\frac{\Delta SALE - \Delta RECT}{AT}$	#34
	Receivables Turnover	$\frac{SALE}{RECT}$	#17
<i>Panel D: Liquidity & Solvency</i>			
	Operating Leverage	$\frac{\Delta SALE - \Delta INVT}{AT}$	N/A
	Cash Cushion	$\frac{CHE}{AT}$	#86
	Book Leverage	$\frac{DLTT + DLC}{SEQ}$	#7
	Δ Current Ratio	$\Delta \left(\frac{ACT}{LCT} \right)$	#11
	Debt Capacity	$\frac{RECT + INVT + PPENT}{AT}$	#71
<i>Panel E: Operating Efficiency</i>			
	Cash Productivity	$\frac{SALE + \Delta RECT}{CHE}$	#76
	Depreciation/PP&E	$\frac{DP}{PPENT}$	#18
	Δ Depreciation	$\frac{\Delta DP}{AT}$	#19
	Tax/Income Ratio	$\frac{TXT}{PI}$	#56
	Asset Turnover	$\frac{SALE}{AT}$	N/A

Notes: This table defines the 25 fundamental signals employed in this study. *Formula* reports the construction using Compustat mnemonics. *Zoo ID* denotes the corresponding characteristic identifier in Feng, Giglio, and Xiu (2020). Signals marked “N/A” are simple accounting-based constructions that are not explicitly mapped to a Zoo ID. Variables are Compustat annual items (fiscal-year end) aligned to the June formation date using the point-in-time rules in Section 2.2.1. See Table 16 for the compustat detailed mnemonics details.

Table 16: Compustat Mnemonics Used in Fundamental Signal Construction

Mnemonic	Description	Mnemonic	Description	Mnemonic	Description
<i>Income statement items</i>					
REVT	Total revenue	SALE	Net sales	COGS	Cost of goods sold
XSGA	SG&A expense	NI	Net income	PI	Pretax income
TXT	Income taxes	DVC	Common dividends		
<i>Balance sheet items</i>					
AT	Total assets	CEQ	Common equity	SEQ	Shareholders' equity (total)
ACT	Current assets	LCT	Current liabilities	WC	Working capital (constructed; see signal definition)
CHE	Cash & short-term investments	RECT	Receivables	INVT	Inventories
PPENT	Net PP&E	PPEGT	Gross PP&E	DLTT	Long-term debt
DLC	Short-term debt (current debt)	CSHO	Common shares outstanding		
<i>Cash flow / non-cash expense items</i>					
CAPX	Capital expenditures	DP	Depreciation & amortization		

Notes: Mnemonics follow WRDS/Compustat Fundamentals Annual conventions. All items are annual (fiscal-year) accounting variables and are aligned to the June formation date using the paper's point-in-time rules. **Notation:** $\Delta X = X_t - X_{t-1}$ denotes the annual change. AT denotes average total assets, typically $(AT_t + AT_{t-1})/2$.

A.2 Signal Universe Diagnostics

A.2.1 Size $\times$ Signal portfolio statistics

Table 17: Portfolio Statistics Summary by Signal (1963–2024)

Category	Signal	Total	Avg/Port	Min	Max	Median	Small%	Big%	Low%	High%
<i>Panel A: Profitability</i>										
	Gross Profitability	1,889	75.6	48	135	65	29.0	14.9	22.3	22.9
	Operating Profitability	1,541	61.7	32	151	58	29.3	14.5	26.1	20.9
	Return on Assets	1,903	76.1	40	176	72	29.0	14.9	24.0	22.3
	Δ Asset Turnover	1,808	72.3	45	126	67	28.5	15.2	20.8	20.9
	Δ Profit Margin	1,790	71.6	46	121	65	28.4	15.2	20.7	21.8
<i>Panel B: Investment</i>										
	Return on Invested Capital	2,118	84.7	39	165	78	29.4	14.7	20.8	23.6
	Asset Growth	1,815	72.6	41	141	66	28.6	15.2	22.1	25.1
	Investment (CapEx)	1,815	72.6	41	141	66	28.6	15.2	22.1	25.1
	CapEx + Inventory Growth	1,555	62.2	37	104	57	28.4	15.5	22.6	22.1
	Net Share Issuance	2,235	89.4	47	142	90	27.6	15.0	23.5	23.5
	CapEx Growth	1,655	66.2	39	113	59	28.3	15.5	20.8	22.6
<i>Panel C: Accruals & Quality</i>										
	Sales Growth	1,801	72.1	48	146	62	28.4	15.2	20.9	25.2
	Working Capital Accruals	1,433	57.3	32	119	55	28.4	15.5	21.7	23.3
	Sales-Receivables Gap	1,745	69.8	47	134	61	28.6	15.1	22.1	23.4
	Receivables Turnover	1,819	72.8	50	147	66	28.9	14.9	24.0	18.9
<i>Panel D: Liquidity & Solvency</i>										
	Operating Leverage	1,728	69.1	48	135	62	28.7	15.2	21.0	24.1
	Cash Cushion	1,882	75.3	47	192	64	29.1	14.9	16.4	30.1
	Book Leverage	2,109	84.4	55	220	72	29.4	14.7	29.0	18.4
	Δ Current Ratio	1,462	58.5	36	114	53	28.2	15.6	23.0	23.2
	Debt Capacity	1,748	69.9	39	148	65	29.6	14.7	26.8	17.2
<i>Panel E: Operating Efficiency</i>										
	Cash Productivity	1,726	69.0	41	170	58	28.7	15.1	29.5	16.7
	Depreciation/PPE	1,712	68.5	44	155	59	29.4	15.0	16.7	25.8
	Δ Depreciation	1,686	67.5	36	123	60	28.9	15.2	23.5	23.6
	Tax/Income Ratio	1,907	76.3	41	153	70	28.9	14.9	22.7	19.8
	Asset Turnover	1,907	76.3	39	133	69	28.9	14.9	22.4	19.6
MEAN		1,791	71.7	42	144	64	28.8	15.1	22.6	22.4

Notes: This table reports time-series averages of portfolio characteristics across 25 fundamental signals. Each signal is independently sorted into quintiles based on NYSE breakpoints. The sample includes U.S. common stocks (1963–2024) that pass the Fama-French NYSE size filter (20th percentile and price $\geq$ \$5) screen. *Total* is the average number of firms across all 25 portfolios (5 size $\times$ 5 signal quintiles). *Avg/Port* is the mean portfolio size. *Min/Max* indicate the smallest and largest portfolios. *Small%/Big%* report the percentage of firms in the small-cap and large-cap quintiles. *Low%/High%* report the percentage of firms in the lowest and highest signal quintiles. Signals are grouped into five categories.

A.2.2 Signal and return correlation structure

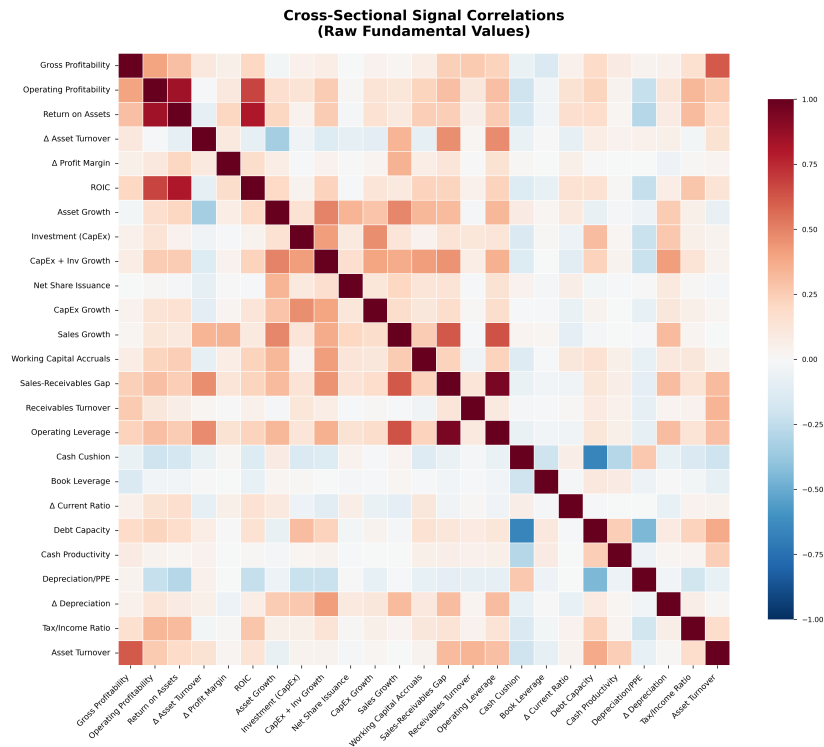


Figure 8: Average Cross-Sectional Spearman Rank Correlations (25 Fundamental Signals). This heatmap displays the pairwise dependency structure of the feature set used for machine learning. The low average correlation (0.091) confirms that the selected signals capture distinct economic dimensions, minimizing multicollinearity. Clusters of higher correlation (e.g., within Profitability) reflect natural economic linkages, while the overall sparsity supports the hypothesis of a broad, heterogeneous information set. Sample period: July 1963–2024.

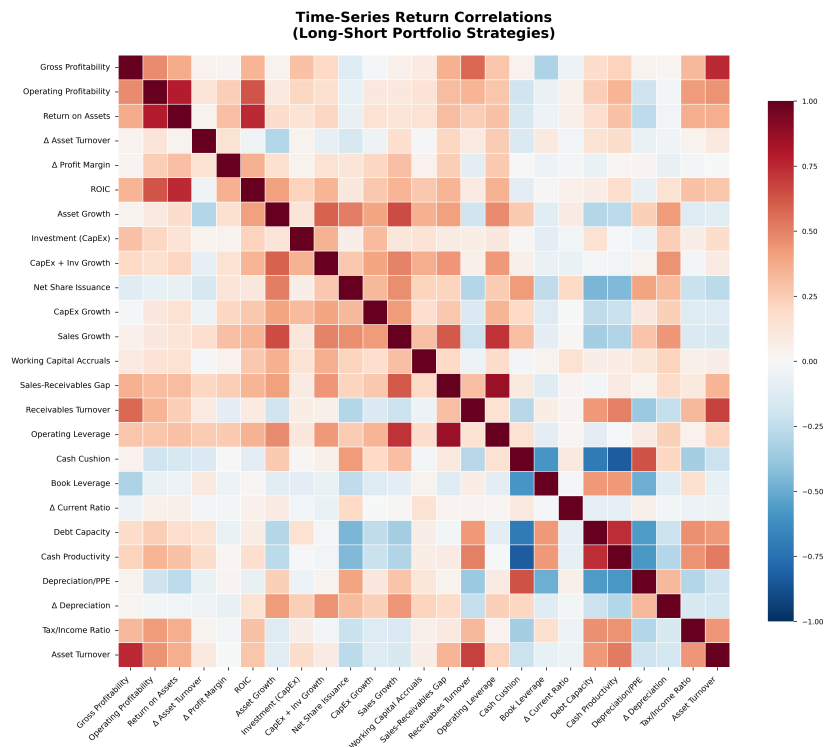


Figure 9: Time-Series Pearson Correlations of Long-Short Portfolio Returns. This figure illustrates the diversification potential available to the dynamic allocation model. Although the final strategy is long-only, the correlation structure of the long-short spreads isolates the idiosyncratic behavior of each factor, stripping out the common market beta. The low average correlation (0.110) and the presence of deep negative correlations (down to -0.830) indicate that the underlying sources of alpha are structurally distinct, providing the necessary breadth for the algorithm to rotate active risk effectively across market regimes. Sample period: July 1963–2024.

A.3 Long-Short Portfolio Construction

Table 18: Long-Short Portfolio Composition (1963–2024)

Category	Signal	Avg P1	Avg P10	Total	P1%	P10%
<i>Panel A: Profitability</i>						
	Gross Profitability	47.0	52.7	99.7	47.2	52.8
	Operating Profitability	53.0	58.0	111.0	47.8	52.2
	Return on Assets	52.3	66.2	118.5	44.2	55.8
	Δ Asset Turnover	49.7	50.2	99.9	49.8	50.2
	Δ Profit Margin	52.6	55.2	107.7	48.8	51.2
<i>Panel B: Investment</i>						
	Return on Invested Capital	49.2	64.3	113.5	43.3	56.7
	Asset Growth	41.0	62.6	103.6	39.6	60.4
	Investment (CapEx)	48.7	54.9	103.5	47.0	53.0
	CapEx + Inventory Growth	37.7	51.7	89.4	42.1	57.9
	Net Share Issuance	53.1	63.0	116.1	45.7	54.3
	CapEx Growth	48.8	58.0	106.8	45.7	54.3
<i>Panel C: Accruals & Quality</i>						
	Sales Growth	41.3	65.8	107.1	38.6	61.4
	Working Capital Accruals	44.8	52.8	97.6	45.9	54.1
	Sales-Receivables Gap	41.0	55.3	96.3	42.6	57.4
	Receivables Turnover	55.2	46.6	101.8	54.2	45.8
<i>Panel D: Liquidity & Solvency</i>						
	Operating Leverage	40.2	58.9	99.1	40.6	59.4
	Cash Cushion	42.0	87.3	129.4	32.5	67.5
	Book Leverage	82.9	39.8	122.7	67.5	32.5
	Δ Current Ratio	56.3	57.7	114.0	49.4	50.6
	Debt Capacity	74.6	44.0	118.5	62.9	37.1
<i>Panel E: Operating Efficiency</i>						
	Cash Productivity	75.7	39.4	115.1	65.8	34.2
	Depreciation/PPE	43.0	68.6	111.5	38.5	61.5
	Δ Depreciation	43.6	61.2	104.8	41.6	58.4
	Tax/Income Ratio	55.8	48.0	103.8	53.8	46.2
	Asset Turnover	55.0	48.9	103.9	52.9	47.1
MEAN		51.4	56.4	107.8	47.5	52.5

Notes: This table reports the average composition of long-short portfolios constructed from decile sorts on 25 fundamental signals. *Avg P1* (*P10*) is the time-series average number of firms in the lowest (highest) decile portfolio. *Total* is the sum of P1 and P10. *P1%* (*P10%*) is the percentage of total firms allocated to each portfolio. Portfolios are formed annually in June using NYSE decile breakpoints applied to the filtered universe (NYSE 20th-percentile market-cap breakpoint filter + price ≥ \$5). Asymmetries in P1 vs. P10 reflect signal-specific data completeness constraints. For example, Book Leverage (67.5% in P1 vs. 32.5% in P10) indicates more firms with high leverage than low leverage, while Cash Cushion shows the opposite pattern (32.5% vs. 67.5%). Sample period: July 1963 to June 2024.

B Basket Construction Details

Table 19: Signal Universe and Basket Classification

Category	Signal Name	Leg	Economic Basket	Zoo ID
<i>Panel A: Profitability (5 signals)</i>				
	Gross Profitability	P10	Defensive	#88
	Operating Profitability	P10	Defensive	#95
	Return on Assets	P10	Defensive	#82
	Δ Asset Turnover	P10	Offensive	#72
	Δ Profit Margin	P10	Defensive	#74
<i>Panel B: Investment (6 signals)</i>				
	Return on Invested Capital	P10	Defensive	#70
	Asset Growth	P1	Offensive	#92
	Investment (CapEx)	P1	Offensive	#53
	CapEx + Inventory Growth	P1	Defensive	#81
	Net Share Issuance	P1	Defensive	#73
	CapEx Growth	P1	Offensive	#63
	Sales Growth	P1	Offensive	#24
<i>Panel C: Accruals & Quality (4 signals)</i>				
	Working Capital Accruals	P1	Defensive	#28
	Sales-Receivables Gap	P10	Defensive	#34
	Receivables Turnover	P10	Offensive	#17
	Operating Leverage	P10	Offensive	N/A
<i>Panel D: Liquidity & Solvency (4 signals)</i>				
	Cash Cushion	P10	Defensive	#86
	Debt Capacity	P10	Defensive	#71
	Book Leverage	P1	Defensive	#7
	Δ Current Ratio	P10	Defensive	#11
<i>Panel E: Operating Efficiency (5 signals)</i>				
	Asset Turnover	P10	Offensive	N/A
	Cash Productivity	P10	Offensive	#76
	Depreciation/PP&E	P1	Defensive	#18
	Δ Depreciation	P1	Defensive	#19
	Tax/Income Ratio	P10	Defensive	#56

Notes: This table presents the 25 fundamental signals employed across all five basket construction strategies. Signals are grouped into five economic categories following the asset pricing literature: Profitability, Investment, Accruals & Quality, Liquidity & Solvency, and Operating Efficiency. *Leg* indicates portfolio direction: P10 = buy top decile (high values are favorable), P1 = buy bottom decile (low values are favorable, contrarian). *Economic Basket* denotes the **fixed allocation** used in Basket 1 (Economic Classification): Offensive (9 signals) captures growth, expansion, and velocity metrics with procyclical behavior; Defensive (16 signals) aggregates quality, profitability, liquidity, and conservatism metrics providing downside protection. For Baskets 2–5 (Dynamic Beta Strategies), signals are **not** pre-assigned to baskets; instead, all 25 signals form a unified pool from which factors are dynamically allocated each month based on time-varying market beta estimates—top 50% highest beta $\rightarrow$ High Beta basket, bottom 50% $\rightarrow$ Low Beta basket. *Zoo ID* references Feng, Giglio, and Xiu (2020), “Taming the Factor Zoo,” *Journal of Finance*. See Appendix A.1 for detailed variable definitions and Compustat mnemonics.

C Machine Learning Implementation Details

C.1 Feature Set

Table 20: Complete Feature Set (35 Features)

Feature	Formula	Window
A. Momentum Features (6)		
momentum_3m	$\frac{1}{50} \sum_{i=1}^{50} \bar{r}_{i,[t-3:t-1]}$	3 months
momentum_6m	$\frac{1}{50} \sum_{i=1}^{50} \bar{r}_{i,[t-6:t-1]}$	6 months
momentum_12m	$\frac{1}{50} \sum_{i=1}^{50} \bar{r}_{i,[t-12:t-1]}$	12 months
momentum_3m_median	$\text{median}(\bar{r}_{i,[t-3:t-1]})$	3 months
momentum_6m_median	$\text{median}(\bar{r}_{i,[t-6:t-1]})$	6 months
momentum_trend_6m	$\frac{1}{50} \sum_{i=1}^{50} [\prod_{k=t-6}^{t-1} (1 + r_{ik}) - 1]$	6 months
B. Dispersion & Volatility Features (8)		
dispersion_5m	$\text{std}_i(\bar{r}_{i,[t-5:t-1]})$	5 months
dispersion_12m	$\text{std}_i(\bar{r}_{i,[t-12:t-1]})$	12 months
downside_vol_6m	$\frac{1}{50} \sum_{i=1}^{50} \text{std}(r_{ij} \mid r_{ij} < 0)_{[t-6:t-1]}$	6 months
upside_vol_6m	$\frac{1}{50} \sum_{i=1}^{50} \text{std}(r_{ij} \mid r_{ij} > 0)_{[t-6:t-1]}$	6 months
vol_asymmetry_6m	$\text{downside_vol_6m} / \text{upside_vol_6m}$	6 months
max_drawdown_12m	$\frac{1}{50} \sum_{i=1}^{50} \min_j \left(\frac{P_{ij}}{\max_{k \leq j} P_{ik}} - 1 \right)$	12 months
avg_vol_3m	$\frac{1}{50} \sum_{i=1}^{50} \sigma_{i,[t-3:t-1]}$	3 months
vol_of_vol_6m	$\frac{1}{50} \sum_{i=1}^{50} \text{std}(\sigma_{i,[t-6:t-1]})$	6 months
C. Beta Features (7)		
beta_range	$\max_j \beta_{j,t-1} - \min_j \beta_{j,t-1}$	Point-in-time
beta_dispersion	$\text{std}_j(\beta_{j,t-1})$	Point-in-time
beta_skewness	$\text{skew}_j(\beta_{j,t-1})$	Point-in-time
beta_mean	$\frac{1}{25} \sum_{j=1}^{25} \beta_{j,t-1}$	Point-in-time
beta_momentum_3m	$\frac{1}{25} \sum_{j=1}^{25} (\beta_{j,t-1} - \beta_{j,t-4})$	3 months
beta_momentum_6m	$\frac{1}{25} \sum_{j=1}^{25} (\beta_{j,t-1} - \beta_{j,t-7})$	6 months
beta_trend_12m	$\frac{1}{25} \sum_{j=1}^{25} \text{slope}(\beta_{j,[t-12:t-1]})$	12 months
D. Cross-Sectional Features (8)		
pct_positive	$\frac{1}{50} \sum_{i=1}^{50} \mathbb{1}(r_{i,t-1} > 0)$	1 month
pct_positive_3m	$\frac{1}{50} \sum_{i=1}^{50} \mathbb{1}(\bar{r}_{i,[t-3:t-1]} > 0)$	3 months
concentration_hhi	$\sum_{i=1}^{50} \left(\frac{ r_{i,t-1} }{\sum_j r_{j,t-1} } \right)^2$	1 month
skewness_cs	$\text{skew}_i(r_{i,t-1})$	1 month
kurtosis_cs	$\text{kurt}_i(r_{i,t-1})$	1 month
tail_ratio	$(P_{90} - P_{50}) / (P_{50} - P_{10})$	1 month
top_bottom_spread	$P_{90}(r_{t-1}) - P_{10}(r_{t-1})$	1 month
median_return	$\text{median}_i(r_{i,t-1})$	1 month
E. Interaction Features (6)		
mom_x_disp	$\text{momentum_6m} \times \text{dispersion_5m}$	Composite
mom_x_beta_range	$\text{momentum_6m} \times \text{beta_range}$	Composite
beta_range_x_disp	$\text{beta_range} \times \text{dispersion_5m}$	Composite
downside_x_beta	$\text{downside_vol_6m} \times \text{beta_dispersion}$	Composite
momentum_sharpe	$\text{momentum_6m} / (\text{dispersion_5m} + 10^{-6})$	Composite
beta_vol_ratio	$\text{beta_dispersion} / (\text{avg_vol_3m} + 10^{-6})$	Composite

Notation: $\bar{r}_{i,[t-k:t-1]}$ denotes the average return of portfolio i over months $[t-k, t-1]$. $\beta_{j,t-1}$ denotes the expanding-window CAPM beta for characteristic j estimated through month $t-1$. All features are lagged by +1 month (see Section 3.6.2.2).

C.2 Model Hyperparameters

Table 21: Hyperparameter Specifications Across All Models

Model	Parameter	Value
Logistic Regression	Penalty	L2 (Ridge)
	Regularization strength (C)	1.0
	Solver	LBFGS
	Max iterations	1000
	Class weight	None
	Random state	42
Random Forest	Number of trees	200
	Max depth	3
	Max features	$\log_2(35) \approx 5$
	Min samples leaf	5
	Class weight	None
	Random state	42
XGBoost	Number of estimators	200
	Max depth	5
	Learning rate (η)	0.1
	Subsample	0.8
	Colsample by tree	0.7
	L1 regularization (α)	0.0
	L2 regularization (λ)	0.1
	Min child weight	1
	Gamma	0.0
	Scale pos weight	$n_{\text{LOW}}/n_{\text{HIGH}} \approx 1.70$
	Tree method	Histogram
	Objective	Binary logistic
	Random state	42
LSTM	Architecture	LSTM(32) + Dropout(0.2) + Dense(1, sigmoid)
	LSTM units	32
	Dropout rate	0.2
	Optimizer	Adam ($\alpha = 0.001$)
	Loss function	Binary crossentropy
	Batch size	32
	Epochs	30
	Early stopping	Patience = 3 (monitor: loss)
	Random state	42
	Window (Expanding)	Variable (min 60 months, max 648 months)
	Window (Rolling 10Y)	Fixed (120 months)

Hyperparameter specifications for all machine learning models. Logistic Regression serves as the linear baseline with L2 regularization. Random Forest uses shallow trees (depth=3) with log2 feature sampling to reduce overfitting. XGBoost employs gradient boosting with histogram-based splits and class-weighted loss (≈ 1.70) to handle imbalanced regimes. LSTM uses a single-layer recurrent architecture with two training strategies: Expanding (variable window, 60–648 months) and Rolling 10-Year (fixed 120-month window). All models use random state 42 for reproducibility.

D Supplementary Results and Robustness

D.1 Full Performance Grid (Gross vs. Net)

Table 22: Complete Performance Metrics: Gross vs. Net Returns (1980–2024)

Gross Returns (Signal Power)								Net Returns (Realizability)							
Basket	Model	Ann.Ret. (%)	CAGR (%)	Sharpe	Sortino	Max DD (%)	Date	Basket	Model	Ann.Ret. (%)	CAGR (%)	Sharpe	Sortino	Max DD (%)	Date
<i>Economic</i>	XGB	15.93	14.78	0.596	0.860	-49.0	2009-02	<i>Economic</i>	XGB	15.54	14.34	0.577	0.832	-49.3	2009-02
	RF	15.56	14.35	0.577	0.841	-49.0	2009-02		RF	15.17	13.91	0.557	0.812	-49.3	2009-02
	Logistic L2	15.49	14.29	0.575	0.834	-49.0	2009-02		Logistic L2	15.10	13.85	0.555	0.806	-49.3	2009-02
	LSTM Rolling 10Y	15.30	14.09	0.568	0.832	-49.0	2009-02		LSTM Rolling 10Y	14.91	13.65	0.548	0.804	-49.3	2009-02
	LSTM Expanding	15.32	14.10	0.566	0.827	-49.0	2009-02		LSTM Expanding	14.93	13.65	0.547	0.798	-49.3	2009-02
<i>Beta Exp.</i>	LSTM Rolling 10Y	16.85	15.88	0.649	0.933	-47.5	2009-02	<i>Beta Exp.</i>	LSTM Rolling 10Y	16.47	15.45	0.631	0.906	-47.8	2009-02
	LSTM Expanding	16.76	15.71	0.635	0.916	-47.5	2009-02		LSTM Expanding	16.38	15.28	0.617	0.889	-47.8	2009-02
	RF	16.65	15.57	0.627	0.916	-47.5	2009-02		RF	16.28	15.14	0.608	0.889	-47.8	2009-02
	Logistic L2	16.39	15.29	0.617	0.885	-47.5	2009-02		Logistic L2	16.01	14.86	0.599	0.858	-47.8	2009-02
	XGB	16.05	14.84	0.591	0.877	-47.5	2009-02		XGB	15.68	14.42	0.573	0.849	-47.8	2009-02
<i>Rolling β</i>	LSTM Rolling 10Y	16.71	15.75	0.646	0.931	-44.8	2009-02	<i>Rolling β</i>	LSTM Rolling 10Y	16.34	15.32	0.627	0.903	-45.1	2009-02
	LSTM Expanding	16.55	15.49	0.628	0.906	-44.8	2009-02		LSTM Expanding	16.17	15.06	0.609	0.878	-45.1	2009-02
	RF	16.57	15.44	0.617	0.908	-44.8	2009-02		RF	16.19	15.01	0.599	0.880	-45.1	2009-02
	Logistic L2	16.20	15.08	0.609	0.875	-44.8	2009-02		Logistic L2	15.82	14.65	0.590	0.848	-45.1	2009-02
	XGB	15.87	14.61	0.578	0.864	-44.8	2009-02		XGB	15.49	14.18	0.560	0.835	-45.1	2009-02
<i>EWMA β</i>	LSTM Rolling 10Y	16.55	15.57	0.638	0.913	-45.8	2009-02	<i>EWMA β</i>	LSTM Rolling 10Y	16.16	15.13	0.618	0.885	-46.1	2009-02
	LSTM Expanding	16.50	15.43	0.624	0.896	-45.8	2009-02		LSTM Expanding	16.12	14.99	0.604	0.869	-46.1	2009-02
	Logistic L2	16.31	15.19	0.612	0.880	-45.8	2009-02		Logistic L2	15.93	14.75	0.593	0.853	-46.1	2009-02
	RF	16.35	15.17	0.604	0.876	-45.8	2009-02		RF	15.96	14.73	0.586	0.850	-46.1	2009-02
	XGB	15.67	14.36	0.566	0.840	-45.8	2009-02		XGB	15.29	13.92	0.547	0.812	-46.1	2009-02
<i>Kalman β</i>	LSTM Rolling 10Y	16.58	15.60	0.639	0.918	-44.9	2009-02	<i>Kalman β</i>	LSTM Rolling 10Y	16.20	15.16	0.620	0.890	-45.2	2009-02
	LSTM Expanding	16.62	15.56	0.629	0.909	-44.9	2009-02		LSTM Expanding	16.24	15.12	0.610	0.881	-45.2	2009-02
	RF	16.50	15.35	0.613	0.897	-44.9	2009-02		RF	16.11	14.91	0.594	0.869	-45.2	2009-02
	Logistic L2	16.26	15.13	0.609	0.878	-44.9	2009-02		Logistic L2	15.87	14.69	0.590	0.850	-45.2	2009-02
	XGB	15.81	14.54	0.575	0.858	-44.9	2009-02		XGB	15.43	14.11	0.557	0.829	-45.2	2009-02
<i>NYSE Benchmark</i>		14.71	13.95	0.613	0.846	-48.4	2009-02	<i>NYSE Benchmark</i>		14.71	13.95	0.613	0.846	-48.4	2009-02

Notes: This table presents the complete performance metrics for all 25 basket-model combinations under gross returns (left panel, signal power) and net returns (right panel, realizability after transaction costs). The sample consists of NYSE/AMEX/NASDAQ stocks with price $\geq$ \$5 and market capitalization $\geq$ 20th percentile (NYSE) from July 1980 to June 2024 (528 months). Within each basket, models are sorted by Sharpe ratio (descending). Net returns incorporate Frazzini (2018) transaction costs applied annually in June. The NYSE Benchmark is an equal-weighted portfolio of all eligible stocks with monthly rebalancing (no transaction costs, identical for both panels). *CAGR* = compound annual growth rate, *Sharpe* = Sharpe ratio, *Sortino* = Sortino ratio, *Max DD* = maximum drawdown, *Date* = date of maximum drawdown. Requires packages `pdfscape` and `booktabs`.

D.2 Turnover and Transaction Costs

Table 23: Portfolio Turnover by Basket Strategy (1980–2024)

Basket Strategy	Panel A: Aggregate			Panel B: By Period								
	Stock	Inter	Total	1980–1999			2000–2009			2010–2024		
	(%)	(%)	(%)	Stock	Inter	Total	Stock	Inter	Total	Stock	Inter	Total
Economic Basket	100.9	0.0	100.9	116.7	0.0	116.7	106.3	0.0	106.3	77.3	0.0	77.3
Beta Expansion	115.9	0.0	115.9	138.1	0.0	138.1	117.3	0.0	117.3	86.8	0.0	86.8
Rolling Beta	124.1	2.5	126.6	137.2	1.3	138.5	143.2	4.0	147.2	94.7	3.2	97.9
EWMA Beta	120.9	5.3	126.2	131.4	5.9	137.3	137.7	4.8	142.5	96.6	4.8	101.4
Kalman Filter Beta	120.9	3.3	124.2	133.1	2.1	135.2	138.8	3.2	142.0	93.6	4.8	98.4

Note: This table reports portfolio turnover for different basket strategies from 1980 to 2024. Panel A shows the aggregate (time-averaged) turnover, while Panel B breaks down turnover by three periods: 1980–1999, 2000–2009, and 2010–2024. Turnover is measured as the annual percentage of the portfolio that is rebalanced. “Stock” refers to stock-level turnover from annual June rebalancing (Fama-French convention). “Inter” refers to inter-basket turnover from switches between Offensive/Defensive or High Beta/Low Beta strategies. All values are in percentage points (%).

D.2.1 Transaction Costs by Basket Strategy

Table 24: Transaction Costs by Basket Strategy (1980–2024)

Basket Strategy	Panel A: Aggregate			Panel B: By Period								
	Stock	Inter	Total	1980–1999			2000–2009			2010–2024		
	(bps)	(bps)	(bps)	Stock	Inter	Total	Stock	Inter	Total	Stock	Inter	Total
Economic	37.0	0.0	37.0	60.7	0.0	60.7	23.7	0.0	23.7	15.8	0.0	15.8
Beta Expansion	37.5	0.0	37.5	62.6	0.0	62.6	23.3	0.0	23.3	15.3	0.0	15.3
Rolling Beta	37.1	0.7	37.7	61.4	0.6	62.0	23.4	0.8	24.2	15.3	0.6	16.0
EWMA Beta	36.6	1.8	38.4	60.4	2.9	63.4	23.0	1.0	24.0	15.3	1.0	16.3
Kalman Beta	37.2	0.9	38.1	62.0	1.1	63.0	23.2	0.6	23.8	15.1	1.0	16.0
<i>Mean</i>	<i>39.5</i>	<i>0.7</i>	<i>40.2</i>	<i>60.8</i>	<i>0.9</i>	<i>61.7</i>	<i>35.4</i>	<i>0.4</i>	<i>35.7</i>	<i>15.4</i>	<i>0.6</i>	<i>16.0</i>

Notes: This table reports annual transaction costs for each basket strategy. *Panel A* presents aggregate costs (July 1980–June 2024, 528 months). *Panel B* decomposes costs by period. *Stock* TC reflects one-way costs at annual rebalancing (June) following Frazzini, Israel, and Moskowitz (2012): 50 bps (1980–1999), 30 bps (2000–2009), 20 bps (2010–2024). *Inter* TC captures switching costs between High/Low Beta portfolios (Baskets 2–5 only; Basket 1 uses Economic cycles, hence Inter = 0). *Total* = Stock + Inter. All costs are annualized and expressed in basis points. The NYSE Benchmark is reported gross (no transaction costs applied).

D.3 Comprehensive Factor Regressions

Table 25: Comprehensive Factor Analysis: Classic Baskets and ML Strategies (Net Returns)

Basket	Strategy	FF5			FF6			FF7			Q5				
		α	t	R^2	α	t	R^2	α	t	R^2	α	t	R^2		
Panel A: Classic Buy&Hold Baskets (Net)															
Economic	(Offensive)	-0.29	-0.23	0.924	1.26		0.96	0.939	1.13		0.79	0.939	2.11	1.31	0.904
	(Defensive)	0.90	0.77	0.943	2.14	*	1.68	0.954	1.95		1.42	0.954	2.22	1.36	0.936
Beta Exp.	(Offensive)	1.39	1.04	0.938	2.82	*	1.94	0.951	2.60	*	1.66	0.951	3.00	1.59	0.934
	(Defensive)	0.01	0.01	0.933	1.28		1.17	0.945	1.15		0.99	0.945	1.80	1.22	0.911
Rolling β	(Offensive)	1.39	0.96	0.932	2.91	*	1.80	0.945	2.70		1.57	0.945	3.25	1.57	0.929
	(Defensive)	-0.11	-0.10	0.929	1.08		1.03	0.939	0.94		0.83	0.940	1.37	0.96	0.903
EWMA β	(Offensive)	1.18	0.84	0.933	2.62	*	1.70	0.945	2.38		1.45	0.946	2.98	1.51	0.932
	(Defensive)	0.16	0.14	0.927	1.33		1.19	0.937	1.21		0.99	0.938	1.64	1.12	0.900
Kalman β	(Offensive)	1.33	0.95	0.933	2.78	*	1.84	0.945	2.49		1.54	0.946	3.03	1.56	0.931
	(Defensive)	-0.14	-0.12	0.927	1.08		0.96	0.938	1.00		0.82	0.939	1.49	0.99	0.901
Panel B: Machine Learning Strategies (Net)															
Economic	Logistic	0.32	0.26	0.933	1.66		1.27	0.945	1.40		0.99	0.945	2.10	1.31	0.923
	RF	-0.28	-0.23	0.926	1.25		0.98	0.941	1.06		0.77	0.942	1.90	1.21	0.909
	XGB	0.25	0.20	0.927	1.73		1.31	0.941	1.54		1.08	0.941	2.14	1.32	0.908
	LSTM-Exp	-0.08	-0.06	0.925	1.42		1.14	0.940	1.17		0.88	0.941	2.03	1.27	0.913
	LSTM-Roll	-0.17	-0.14	0.927	1.31		1.06	0.941	1.09		0.82	0.942	1.90	1.19	0.915
Beta Exp.	Logistic	0.37	0.31	0.931	1.86	*	1.66	0.946	1.56		1.26	0.946	1.84	1.25	0.916
	RF	1.49	1.12	0.938	2.77	*	1.94	0.948	2.53	*	1.66	0.949	2.77	1.49	0.932
	XGB	1.10	0.78	0.935	2.53	*	1.66	0.947	2.26		1.36	0.948	2.67	1.32	0.930
	LSTM-Exp	0.90	0.83	0.947	2.00	*	1.80	0.955	1.76		1.46	0.956	1.95	1.55	0.936
	LSTM-Roll	1.12	1.03	0.947	2.16	*	1.92	0.954	2.06	*	1.71	0.954	2.28	*	1.74
Rolling β	Logistic	0.23	0.18	0.918	1.77		1.52	0.934	1.45		1.12	0.934	1.42	0.93	0.902
	RF	1.80	1.26	0.929	3.14	**	2.02	0.940	2.85	*	1.71	0.941	2.99	1.49	0.926
	XGB	1.30	0.87	0.926	2.85	*	1.74	0.941	2.55		1.44	0.942	2.84	1.30	0.924
	LSTM-Exp	1.16	0.98	0.941	2.21	*	1.83	0.948	1.94		1.48	0.948	1.75	1.39	0.933
	LSTM-Roll	1.35	1.14	0.940	2.39	**	1.99	0.947	2.24	*	1.71	0.948	2.09	1.59	0.931
EWMA β	Logistic	0.28	0.21	0.915	1.84		1.54	0.931	1.49		1.12	0.932	1.36	0.87	0.899
	RF	1.56	1.09	0.929	2.80	*	1.83	0.938	2.47		1.51	0.939	2.58	1.31	0.927
	XGB	1.06	0.72	0.926	2.54		1.60	0.940	2.17		1.26	0.941	2.45	1.15	0.925
	LSTM-Exp	1.12	0.88	0.937	2.17	*	1.65	0.944	1.86		1.30	0.945	1.52	1.15	0.930
	LSTM-Roll	1.23	0.98	0.936	2.29	*	1.76	0.944	2.15		1.52	0.944	1.81	1.32	0.928
Kalman β	Logistic	0.33	0.25	0.919	1.82		1.51	0.933	1.49		1.11	0.934	1.47	0.94	0.903
	RF	1.56	1.12	0.930	2.82	*	1.91	0.940	2.46		1.55	0.941	2.59	1.35	0.927
	XGB	1.05	0.72	0.927	2.52		1.61	0.940	2.12		1.25	0.942	2.40	1.14	0.925
	LSTM-Exp	1.18	0.98	0.939	2.23	*	1.81	0.946	1.91		1.41	0.947	1.73	1.32	0.931
	LSTM-Roll	1.14	0.94	0.938	2.17	*	1.73	0.945	2.02		1.48	0.946	1.86	1.35	0.929

Notes: This table reports risk-adjusted alphas (α , in % per annum), t -statistics (Newey-West HAC, 12 lags), and R^2 for all ML strategies across five baskets and four factor models. All returns are net of transaction costs. FF5/FF6/FF7 are Fama–French factor models: FF5 includes market (MKT), size (SMB), value (HML), profitability (RMW), and investment (CMA); FF6 extends FF5 by adding momentum (UMD); and FF7 extends FF6 by adding quality-minus-junk (QMJ). Q5 refers to the Hou–Xue–Zhang q -factor model (market, size, investment, profitability, and expected growth). Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sample: July 1980–June 2024.

D.4 Incremental Alpha Tests

D.4.1 All ML Strategies vs 50/50 Benchmark

Table 26: Incremental Alpha: All ML Strategies vs 50/50 Benchmark (Full)

Basket	Model	FF6				Q5			
		$\Delta\alpha$	t	p	q (FDR)	$\Delta\alpha$	t	p	q (FDR)
Economic	Logistic	-0.04	-0.14	0.890	0.927	-0.06	-0.21	0.830	0.899
	RF	-0.45	-1.54	0.124	0.636	-0.26	-0.83	0.408	0.899
	XGB	0.03	0.12	0.907	0.927	-0.03	-0.10	0.918	0.918
	LSTM-Exp	-0.28	-0.88	0.378	0.859	-0.13	-0.46	0.649	0.899
	LSTM-Roll	-0.39	-1.23	0.219	0.781	-0.26	-0.88	0.377	0.899
Beta Exp.	Logistic	-0.19	-0.36	0.721	0.927	-0.56	-0.83	0.406	0.899
	RF	0.72	1.58	0.114	0.636	0.37	0.71	0.477	0.899
	XGB	0.48	1.04	0.300	0.825	0.27	0.49	0.624	0.899
	LSTM-Exp	-0.05	-0.09	0.927	0.927	-0.45	-0.65	0.517	0.899
	LSTM-Roll	0.10	0.20	0.844	0.927	-0.12	-0.17	0.863	0.899
Rolling β	Logistic	-0.23	-0.33	0.740	0.927	-0.89	-1.00	0.319	0.899
	RF	1.14	*	1.92	0.054	0.68	0.90	0.370	0.899
	XGB	0.86	1.44	0.151	0.636	0.53	0.66	0.508	0.899
	LSTM-Exp	0.21	0.30	0.761	0.927	-0.56	-0.62	0.534	0.899
	LSTM-Roll	0.40	0.58	0.564	0.927	-0.22	-0.25	0.803	0.899
EWMA β	Logistic	-0.13	-0.21	0.836	0.927	-0.95	-1.13	0.257	0.899
	RF	0.82	1.43	0.153	0.636	0.27	0.38	0.705	0.899
	XGB	0.57	0.97	0.330	0.825	0.14	0.18	0.860	0.899
	LSTM-Exp	0.20	0.29	0.772	0.927	-0.79	-0.95	0.341	0.899
	LSTM-Roll	0.32	0.48	0.632	0.927	-0.50	-0.64	0.520	0.899
Kalman β	Logistic	-0.11	-0.18	0.854	0.927	-0.79	-0.97	0.330	0.899
	RF	0.89	1.61	0.107	0.636	0.33	0.46	0.642	0.899
	XGB	0.59	1.04	0.297	0.825	0.14	0.19	0.852	0.899
	LSTM-Exp	0.29	0.45	0.651	0.927	-0.53	-0.64	0.521	0.899
	LSTM-Roll	0.24	0.37	0.710	0.927	-0.40	-0.51	0.611	0.899

Notes: This table reports incremental alphas ($\Delta\alpha$, annualized in % p.a.) for all 25 ML strategies (5 baskets $\times$ 5 models) against a 50/50 benchmark (50% Buy&Hold Offensive + 50% Defensive). Incremental alphas are computed by regressing the difference between ML and benchmark returns on Fama–French 6-factor (FF6) and Hou–Xue–Zhang q -factor (Q5) models. t -statistics use Newey–West HAC standard errors (12 lags) on monthly data. p = raw p -value; q (FDR) = Benjamini–Hochberg FDR-adjusted p -value ($\alpha = 10\%$). All returns are net of transaction costs. Sample: July 1980–June 2024. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D.4.2 ML vs Offensive and Defensive Benchmarks (FF6)

Table 27: Incremental Alpha Robustness: ML vs Offensive and Defensive Benchmarks (FF6)

Basket	Model	vs Offensive				vs Defensive			
		$\Delta\alpha$		t	p	$\Delta\alpha$		t	p
Economic	Logistic	0.40	*	1.65	0.099	-0.47		-1.07	0.284
	RF	-0.01		-0.06	0.949	-0.89	*	-1.83	0.067
	XGB	0.46	*	1.71	0.087	-0.41		-1.06	0.290
	LSTM-Exp	0.15		0.55	0.580	-0.72		-1.48	0.138
	LSTM-Roll	0.04		0.14	0.887	-0.83	*	-1.80	0.072
Beta Exp.	Logistic	-0.96		-1.41	0.159	0.58		0.93	0.351
	RF	-0.05		-0.13	0.895	1.49	**	2.05	0.041
	XGB	-0.29		-0.71	0.476	1.25	*	1.69	0.091
	LSTM-Exp	-0.82		-1.33	0.184	0.72		1.07	0.287
	LSTM-Roll	-0.67		-1.06	0.288	0.87		1.32	0.187
Rolling β	Logistic	-1.14		-1.21	0.228	0.68		0.82	0.413
	RF	0.23		0.48	0.632	2.06	*	1.94	0.052
	XGB	-0.06		-0.10	0.922	1.77	*	1.76	0.078
	LSTM-Exp	-0.70		-0.90	0.370	1.13		1.12	0.264
	LSTM-Roll	-0.51		-0.64	0.520	1.31		1.34	0.182
EWMA β	Logistic	-0.78		-0.93	0.355	0.51		0.63	0.527
	RF	0.18		0.39	0.693	1.47		1.46	0.145
	XGB	-0.08		-0.14	0.892	1.21		1.28	0.200
	LSTM-Exp	-0.45		-0.71	0.479	0.84		0.81	0.418
	LSTM-Roll	-0.33		-0.46	0.643	0.96		0.99	0.324
Kalman β	Logistic	-0.96		-1.26	0.207	0.73		0.87	0.384
	RF	0.04		0.09	0.925	1.74	*	1.80	0.072
	XGB	-0.26		-0.44	0.659	1.44		1.62	0.106
	LSTM-Exp	-0.56		-0.86	0.392	1.14		1.19	0.233
	LSTM-Roll	-0.61		-0.88	0.380	1.09		1.19	0.233

Notes: This table reports incremental alphas ($\Delta\alpha$, annualized in % p.a.) for all 25 ML strategies (5 baskets $\times$ 5 models) against two alternative benchmarks: Buy&Hold Offensive (left panel) and Buy&Hold Defensive (right panel). Incremental alphas are computed by regressing the difference between ML and benchmark returns on the Fama–French 6-factor (FF6) model. t -statistics use Newey–West HAC standard errors (12 lags) on monthly data. p = raw p -value. All returns are net of transaction costs. This robustness check complements the main 50/50 benchmark analysis (Table 26). Sample: July 1980–June 2024. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 27 shows that incremental alpha is benchmark-dependent: relative performance is generally weaker and less consistent versus the Offensive benchmark, but becomes economically larger and more statistically pronounced versus the Defensive benchmark, particularly for the Beta Expansion and Kalman baskets under XGBoost.

D.4.3 Alternative Factor Models (CAPM, FF5, FF7)

Table 28: Incremental Alpha: Alternative Factor Models (CAPM, FF5, FF7)

Basket	Model	CAPM			FF5			FF7		
		$\Delta\alpha$	t	p	$\Delta\alpha$	t	p	$\Delta\alpha$	t	p
Economic	Logistic	-0.25	-0.99	0.325	0.02	0.06	0.949	-0.13	-0.48	0.634
	RF	-0.29	-0.90	0.370	-0.58	*	-1.87	0.062	-0.48	0.108
	XGB	0.28	0.94	0.349	-0.05	-0.21	0.832	0.00	0.01	0.991
	LSTM-Exp	-0.45	-1.32	0.188	-0.38	-1.10	0.271	-0.37	-1.14	0.253
	LSTM-Roll	-0.62	*	-1.84	0.065	-0.48	-1.35	0.177	-0.45	0.172
Beta Exp.	Logistic	-0.16	-0.40	0.688	-0.33	-0.64	0.521	-0.32	-0.58	0.564
	RF	0.30	0.65	0.519	0.78	*	1.69	0.090	0.66	0.173
	XGB	-0.44	-1.01	0.310	0.40	0.90	0.367	0.38	0.77	0.441
	LSTM-Exp	0.35	0.71	0.476	0.19	0.37	0.715	-0.12	-0.22	0.825
	LSTM-Roll	0.61	1.29	0.198	0.41	0.75	0.453	0.18	0.33	0.743
Rolling β	Logistic	-0.12	-0.21	0.837	-0.41	-0.59	0.558	-0.37	-0.53	0.599
	RF	0.51	0.83	0.405	1.15	*	1.87	0.062	1.02	0.116
	XGB	-0.36	-0.62	0.536	0.66	1.13	0.260	0.73	1.10	0.271
	LSTM-Exp	0.58	0.85	0.396	0.51	0.71	0.475	0.12	0.16	0.873
	LSTM-Roll	0.83	1.27	0.204	0.71	0.98	0.329	0.42	0.58	0.563
EWMA β	Logistic	-0.06	-0.11	0.915	-0.39	-0.57	0.570	-0.31	-0.48	0.635
	RF	0.25	0.38	0.702	0.89	1.45	0.148	0.68	1.08	0.280
	XGB	-0.70	-1.09	0.278	0.39	0.67	0.501	0.38	0.59	0.552
	LSTM-Exp	0.58	0.74	0.462	0.45	0.62	0.536	0.06	0.09	0.927
	LSTM-Roll	0.77	1.04	0.300	0.57	0.77	0.440	0.36	0.52	0.602
Kalman β	Logistic	-0.07	-0.12	0.906	-0.27	-0.41	0.680	-0.26	-0.40	0.693
	RF	0.37	0.61	0.542	0.96	1.63	0.103	0.71	1.18	0.238
	XGB	-0.52	-0.85	0.393	0.45	0.80	0.421	0.38	0.61	0.540
	LSTM-Exp	0.63	0.91	0.362	0.58	0.86	0.392	0.16	0.24	0.808
	LSTM-Roll	0.70	1.05	0.292	0.54	0.79	0.430	0.28	0.42	0.673

Notes: This table reports incremental alphas ($\Delta\alpha$, annualized in % p.a.) for all 25 ML strategies (5 baskets $\times$ 5 models) against the 50/50 benchmark, using alternative factor models: CAPM (single-factor), Fama–French 5-factor (FF5), and Fama–French 7-factor (FF7, which extends FF6 by adding the quality-minus-junk factor). Incremental alphas are computed by regressing the difference between ML and 50/50 benchmark returns on each factor model. t -statistics use Newey–West HAC standard errors (12 lags) on monthly data. p = raw p -value. All returns are net of transaction costs. This robustness check complements the main FF6/Q5 analysis (Table 26). Sample: July 1980–June 2024. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D.5 Subperiod Robustness

Table 29: Subperiod Robustness: All ML Strategies vs 50/50 Benchmark (FF6)

Basket	Model	P1: 1980–1999		P2: 2000–2009			P3: 2010–2024	
		Sharpe	$\Delta\alpha$		Sharpe	$\Delta\alpha$	Sharpe	$\Delta\alpha$
Economic	Logistic	0.90	0.43		0.54	-0.26	0.75	-0.44
	RF	0.87	0.04		0.57	-1.42 **	0.74	-0.03
	XGB	0.90	0.63		0.61	-0.35	0.74	-0.31
	LSTM-Exp	0.87	0.01		0.54	-0.15	0.75	-0.19
	LSTM-Roll	0.88	-0.03		0.54	-0.15	0.74	-0.64
Beta Exp.	Logistic	0.93	0.84		0.55	-1.53	0.80	0.59
	RF	0.92	1.33 ***		0.61	2.23 **	0.80	0.40
	XGB	0.92	1.10 **		0.52	2.36 **	0.77	-0.61
	LSTM-Exp	0.91	1.23 ***		0.67	-0.27	0.80	0.50
	LSTM-Roll	0.94	1.12 *		0.67	-0.27	0.80	0.58
Rolling β	Logistic	0.92	0.64		0.57	-1.86	0.78	0.83
	RF	0.89	1.18 *		0.65	3.56 ***	0.79	0.82
	XGB	0.90	1.44 **		0.53	3.06 ***	0.74	-0.53
	LSTM-Exp	0.88	1.25 **		0.70	0.10	0.78	0.73
	LSTM-Roll	0.94	1.73 **		0.70	0.10	0.77	0.33
EWMA β	Logistic	0.94	0.59		0.57	-1.83	0.77	0.54
	RF	0.88	0.76		0.63	3.65 ***	0.77	0.29
	XGB	0.91	1.07		0.49	2.79 **	0.73	-1.01
	LSTM-Exp	0.88	0.79		0.71	0.23	0.77	0.45
	LSTM-Roll	0.93	1.31 *		0.71	0.23	0.75	-0.26
Kalman β	Logistic	0.92	0.58		0.58	-1.37	0.78	0.80
	RF	0.88	0.98		0.65	3.56 ***	0.78	0.68
	XGB	0.91	1.37 **		0.52	2.84 ***	0.73	-0.80
	LSTM-Exp	0.89	1.25 **		0.70	0.26	0.78	0.74
	LSTM-Roll	0.93	1.36 **		0.70	0.26	0.76	-0.07

Notes: This table reports subperiod performance metrics for all 25 ML strategies (5 baskets $\times$ 5 models) relative to the 50/50 benchmark (50% Buy&Hold Offensive + 50% Buy&Hold Defensive). Sharpe denotes the net Sharpe ratio of the ML strategy within each subperiod. $\Delta\alpha$ denotes the incremental alpha (in % p.a.) from the FF6 regression of monthly return differentials (ML minus 50/50). Subperiods are: P1 (1980–1999), P2 (2000–2009), and P3 (2010–2024). Statistical significance stars are based on Newey–West HAC t -statistics with 12 lags (not reported). All returns are net of transaction costs. This table complements the main subperiod analysis (Table 11) by reporting results for all models. Sample: July 1980–June 2024. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D.6 Subperiod Incremental Alpha: Full Set of ML Strategies vs. 50/50 Benchmark

Table 30: Subperiod Incremental Alpha: Full Set of ML Strategies vs. 50/50 Benchmark (FF5, FF6, Q5, FF7, Net Returns)

Basket	Model	P1 (1980–1999)							P2 (2000–2009)							P3 (2010–2024)									
		FF5		FF6		FF7		Q5	FF5		FF6		FF7		Q5	FF5		FF6		FF7		Q5			
Economic	Logistic	0.42		0.43		0.29		0.76		-0.26		-0.26		-0.31		-0.55		-0.40		-0.44		-0.50		-0.47	
	RF	-0.30		0.04		0.02		0.57		-1.43	**	-1.42	**	-1.46	***	-0.68		-0.02		-0.03		-0.04		0.09	
	XGB	0.36		0.63		0.51		0.77		-0.36		-0.35		-0.42		0.26		-0.30		-0.31		-0.26		-0.26	
	LSTM-Exp	-0.28		0.01		0.01		0.51		-0.15		-0.15		-0.25		0.04		-0.16		-0.19		-0.29		-0.21	
	LSTM-Roll	-0.21		-0.03		-0.00		0.40		-0.15		-0.15		-0.25		0.04		-0.58		-0.64		-0.75		-0.61	
Beta Exp.	Logistic	0.02		0.84		0.60		-0.19		-1.53		-1.53		-1.56		-1.47		0.52		0.59		0.49		0.51	
	RF	0.89		1.33	***	1.51	***	0.23		2.24	**	2.23	**	2.27	**	1.04		0.31		0.40		0.29		0.15	
	XGB	0.69		1.10	**	1.16	**	0.16		2.36	**	2.36	**	2.46	***	1.24		-0.62		-0.61		-0.78		-0.86	
	LSTM-Exp	0.66	*	1.23	***	1.44	***	0.51		-0.24		-0.27		-0.31		-1.60		0.42		0.50		0.45		0.43	
	LSTM-Roll	1.00	**	1.12	*	1.35	**	0.72		-0.24		-0.27		-0.31		-1.60		0.60		0.58		0.81		0.77	
Rolling β	Logistic	-0.12		0.64		0.39		-0.57		-1.87		-1.86		-1.81		-1.40		0.78		0.83		0.58		0.64	
	RF	0.90		1.18	*	1.36	*	-0.30		3.57	***	3.56	***	3.65	***	2.14	**	0.67		0.82		0.53		0.51	
	XGB	1.12	*	1.44	**	1.49	*	0.12		3.04	**	3.06	***	3.25	***	1.83	**	-0.61		-0.53		-0.84		-0.90	
	LSTM-Exp	0.83		1.25	**	1.54	**	0.39		0.13		0.10		0.21		-1.53		0.67		0.73		0.46		0.54	
	LSTM-Roll	1.50	**	1.73	**	2.05	***	1.14		0.13		0.10		0.21		-1.53		0.27		0.33		0.26		0.27	
EWMA β	Logistic	0.04		0.59		0.35		-0.61		-1.86		-1.83		-1.76		-0.95		0.46		0.54		0.18		0.21	
	RF	0.58		0.76		0.90		-0.71		3.66	***	3.65	***	3.77	***	1.94	**	0.12		0.29		-0.07		-0.14	
	XGB	0.92		1.07		1.05		-0.35		2.78	**	2.79	**	2.96	***	1.33		-1.13		-1.01		-1.39	*	-1.47	*
	LSTM-Exp	0.53		0.79		1.04		-0.09		0.24		0.23		0.36		-1.27		0.35		0.45		0.08		0.11	
	LSTM-Roll	1.16	*	1.31	*	1.59	**	0.71		0.24		0.23		0.36		-1.27		-0.30		-0.26		-0.27		-0.33	
Kalman β	Logistic	-0.05		0.58		0.39		-0.53		-1.39		-1.37		-1.25		-0.93		0.78		0.80		0.50		0.52	
	RF	0.70		0.98		1.06		-0.65		3.58	***	3.56	***	3.60	***	1.76	**	0.55		0.68		0.34		0.27	
	XGB	1.11	*	1.37	**	1.28		-0.20		2.82	**	2.84	***	2.95	***	1.26		-0.86		-0.80		-1.17		-1.26	
	LSTM-Exp	0.83		1.25	**	1.42	**	0.24		0.28		0.26		0.39		-1.26		0.71		0.74		0.43		0.48	
	LSTM-Roll	1.28	**	1.36	**	1.61	**	0.80		0.28		0.26		0.39		-1.26		-0.08		-0.07		-0.06		-0.12	

Notes: This table reports subperiod incremental alphas ($\Delta\alpha$, annualized in % p.a.) for the full set of 25 ML regime-switching strategies (5 baskets $\times$ 5 models) relative to a fixed 50/50 benchmark (50% Buy&Hold Offensive + 50% Buy&Hold Defensive), using net-of-cost returns. For each subperiod *P1* (1980–1999), *P2* (2000–2009), and *P3* (2010–2024), $\Delta\alpha$ is estimated from time-series regressions of (ML minus benchmark) returns on the Fama–French FF5/FF6/FF7 factor models and on the Hou–Xue–Zhang *q*-factor model (Q5). FF7 extends FF6 by adding the quality-minus-junk (QMJ) factor. Stars indicate statistical significance based on Newey–West HAC standard errors (12 lags) computed on monthly data (test statistics not reported in the table). The set of strategies is fixed ex-ante (no re-optimization within subperiods). Sample: July 1980–June 2024. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

D.7 Break-Even Transaction Costs

Table 31: Break-Even Transaction Cost Analysis: All ML Strategies

Basket	Model	NET Return (% p.a.)	Panel A: vs Market (14.71%)			Panel B: vs 50/50 Benchmark		
			Headroom (bps/yr)	Total TC @ BE (bps/yr)	TC Mult.	Headroom (bps/yr)	Total TC @ BE (bps/yr)	TC Mult.
Economic	LSTM-Exp	14.93	22	77	1.41	-77	0	0.00
	LSTM-Roll	14.91	20	74	1.37	-86	0	0.00
	Logistic	15.10	39	94	1.72	-65	0	0.00
	RF	15.17	46	100	1.84	-60	0	0.00
	XGBoost	15.54	83	138	2.53	-25	32	0.56
Beta Expansion	LSTM-Exp	16.38	167	231	3.63	-0	65	1.00
	LSTM-Roll	16.47	176	240	3.77	0	65	1.00
	Logistic	16.01	130	194	3.04	-62	3	0.05
	RF	16.28	157	221	3.46	6	71	1.09
	XGBoost	15.68	97	160	2.52	-10	55	0.85
Rolling Beta	LSTM-Exp	16.17	146	224	2.87	2	80	1.02
	LSTM-Roll	16.34	163	241	3.08	-50	28	0.36
	Logistic	15.82	111	189	2.42	-69	10	0.12
	RF	16.19	148	226	2.90	-13	65	0.83
	XGBoost	15.49	78	156	2.00	-50	28	0.36
EWMA Beta	LSTM-Exp	16.12	141	229	2.59	18	107	1.20
	LSTM-Roll	16.16	145	234	2.65	-29	60	0.68
	Logistic	15.93	122	210	2.38	-54	35	0.39
	RF	15.96	125	214	2.42	-12	77	0.86
	XGBoost	15.29	58	146	1.65	-33	56	0.63
Kalman Beta	LSTM-Exp	16.24	153	224	3.13	19	87	1.28
	LSTM-Roll	16.20	149	220	3.07	-27	41	0.61
	Logistic	15.87	116	188	2.63	-52	16	0.24
	RF	16.11	140	212	2.96	-1	67	0.98
	XGBoost	15.43	72	144	2.01	-20	48	0.71

Notes:

- The transaction-cost figures reported elsewhere reflect one-way stock trading costs only (June rebalance), whereas Real TC here includes both stock-level costs and inter-basket switching costs, hence higher annualized totals.
- NET Return:** Annual return after real transaction costs (stock TC + inter-basket TC).
- Real TC:** Actual transaction costs incurred (56-89 bps/year), including both stock-level rebalancing (June only) and inter-basket switching costs (monthly), computed using Frazzini et al. (2018) schedule.
- Panel A (vs Market):** Break-even analysis vs NYSE Benchmark (14.71% Ann. Return, consistent with Table 1).
- Panel B (vs 50/50):** Break-even analysis vs 50/50 Buy&Hold Offensive/Defensive benchmark, computed net using its own turnover-based TC model (annual June rebalance).
- Headroom:** Additional TC (bps/year) that would eliminate excess return vs benchmark. Negative headroom indicates under-performance.
- Total TC @ BE:** $\max(0, \text{Real TC} + \text{Headroom})$ — maximum sustainable TC level before strategy loses its edge.
- TC Mult.:** $\text{Total TC @ BE} / \text{Real TC}$ (how many times current TC can be sustained).
- When $\text{Real TC} + \text{Headroom} \leq 0$, Total TC @ BE is set to 0 and TC Mult. to 0.00 (strategy already underperforms the benchmark net of current costs).
- TC Multiplier > 1 indicates robustness; $= 0$ indicates the strategy does not beat the benchmark.

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AI Statement

We used third-party generative AI tools (OpenAI ChatGPT, Anthropic Claude, and Google Gemini) for language editing and $\LaTeX$ formatting support. All research design choices, empirical analyses, results, and interpretations were conducted and verified by the author, who take full responsibility for the manuscript.